








Modelling Financial Markets

Chapter 14: Deep Learning and Time-Series Foundation Models



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Can Deep Networks Forecast Better than the Classical Models?

- **Question**: do deep networks forecast better than the classical models of Chapters 5–9?
  - Targets: **returns**, **volatility** and **tail risk**
  - **Deep learning**: models built from many layers of simple non-linear units, trained on data
  - Time-series **foundation models** (since 2024): networks pretrained on billions of observations from other series, used with no estimation on the target series
  - Why test them: their pretraining corpora are dominated by non-financial series (energy, transport, weather, web)
- Answered with the evaluation tools of Chapter 8
  - Strong benchmarks, loss differentials, backtests
- Part I: networks for sequences, attention, foundation models, Test 1 (returns)
- Part II: Test 2, Test 3, case study, artificial intelligence (AI) for scientific discovery
  - Test 2: realised volatility
  - Test 3: left-tail risk (value-at-risk, VaR 1%, and expected shortfall, ES 2.5%)

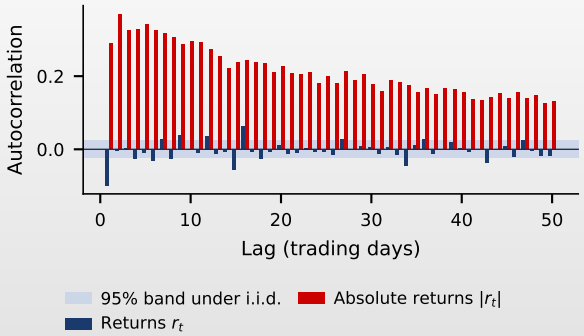
What You Should Be Able to Do

- Explain vanishing gradients as impulse responses of a vector autoregression (VAR)
- Explain the cell state of the LSTM as a first-order autoregression, AR(1)
  - LSTM: long short-term memory, a gated recurrent network
- Turn a zero-shot quantile grid into point, volatility and VaR/ES forecasts
  - Zero-shot: produced without any training on the target series
- Choose the right benchmark for each target
  - The zero forecast for returns
  - Log-HAR for volatility
  - FHS for the tails
- Test a new model correctly: nesting, multiplicity, calibration, contamination and power
- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide

Three Forecasting Targets

- **Returns** $r_t = 100 \ln(P_t / P_{t-1})$ (conditional mean), P_t the close on day t
  - Very low signal-to-noise ratio: predictable patterns are traded away (Chapter 2)
  - Benchmark: the zero forecast
- **Volatility** (conditional variance): persistent and clustered (Chapters 1 and 5), a signal to learn
  - Benchmark: log-HAR, log realised variance regressed on its daily, weekly and monthly averages (heterogeneous autoregressive, Test 2)
- **Tail risk**: the shape of the lower tail, not only the centre (Chapter 7)
  - VaR 1% (value-at-risk): the daily loss exceeded with probability 1%
  - ES 2.5% (expected shortfall): the average loss on the worst 2.5% of days
  - Benchmark: FHS (filtered historical simulation)
  - FHS: the empirical quantile of returns standardised by a GARCH volatility (generalised autoregressive conditional heteroskedasticity, Chapter 5), times the forecast volatility

Where Is the Signal?



- S&P 500 daily returns since 2000 (6 718 days), lags 1–50
  - Light-blue band: $\pm 1.96/\sqrt{T} = \pm 0.024$ (T: number of days), the 95% range of an autocorrelation under i.i.d. returns (independent and identically distributed)
- Returns: lag-1 autocorrelation -0.099
  - 15 of 50 lags outside the band, small and of changing sign
- Absolute returns: 0.29 at lag 1, 0.13 at lag 50
  - All 50 lags outside the band: persistent volatility clustering
- Flexible models can find the volatility signal, not much in the mean

MFM_ch14_acf_signal

From the Multilayer Perceptron to Deep Learning (1/3)

- **Question of the section**: how can a network use information from distant days?
  - And why does a simple recurrent network fail?
- MLP (multilayer perceptron, Chapter 13)

$$h^{(\ell)} = \varphi\left(W^{(\ell)} h^{(\ell-\mathbf{1})} + b^{(\ell)}\right), \qquad h^{(\mathbf{0})} = x$$

  - Each layer applies a non-linear function to a linear combination of the previous layer
  - x: input vector; $h^{(\ell)}$: output of layer ℓ
  - $W^{(\ell)}$, $b^{(\ell)}$: weight matrix and bias vector of layer ℓ, estimated from the data
  - φ: activation, applied element by element: ReLU (rectified linear unit) $\max(0, z)$, tanh, or the sigmoid $\sigma(z) = 1/(1 + e^{-z}) \in (0, 1)$
- An MLP takes a fixed window of inputs
  - A sequence network must carry information from one step to the next

From the Multilayer Perceptron to Deep Learning (2/3)

- Training: backpropagation (Rumelhart, Hinton & Williams (1986)) computes the gradient of the loss with respect to every weight
- Update rule: Adam (adaptive moment estimation, Kingma & Ba (2015))
  - A gradient step normalised by exponential averages of past gradients and of their squares
- Loss for point forecasts: the mean squared error (MSE), minimised by the conditional mean
- Loss for a quantile of level u: the pinball loss

$$\rho_u(y, q) = (u - 1\{y < q\})(y - q)$$

  - y: realised value; q: quantile forecast
  - Asymmetric: values above q cost u per unit, values below cost $1 - u$
  - The expected loss is minimised by the true u-quantile

From the Multilayer Perceptron to Deep Learning (3/3)

- Universal approximation (Hornik, Stinchcombe & White (1989))
  - A wide enough network approximates any continuous function
  - Silent on the sample size needed, the binding constraint with one financial series
  - Regularisation: early stopping, weight decay (a penalty on the size of the weights), ensembles over seeds
  - The validation block must follow the training block
- 2024 Nobel Prize in Physics to Hopfield and Hinton
  - The methods: LeCun, Bengio & Hinton (2015)
  - In finance, Gu, Kelly & Xiu (2020) gain on a cross-section of thousands of stocks, not on one index
  - Fischer & Krauss (2018) apply LSTMs to S&P 500 constituents, forecasting which stocks beat the next-day cross-sectional median



Geoffrey Hinton, Nobel laureate 2024

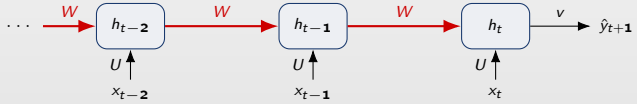
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RNN and LSTM (1/2)

- **Recurrent neural network** (RNN): a hidden state is updated at every step

$$h_t = \tanh\left(W h_{t-1} + U x_t + b\right), \qquad \hat{y}_{t+1} = v^\top h_t + c$$

- The update uses its previous value and the new input
- h_t: hidden state, the memory of the network at t; x_t: input at t (e.g. today's return); $\hat{y}_{t+1}$: forecast
- W, U, b: recurrent weights, input weights and bias; v, c: output weights and intercept; tanh keeps h_t in $(-1, 1)$
- The same weights at every step: unrolled over time, a very deep network with shared weights



RNN and LSTM (2/2)

- **Long short-term memory** (LSTM, Hochreiter & Schmidhuber (1997)): gates decide what to forget, what to write and what to show

$$f_t, i_t, o_t = \sigma(W_\bullet x_t + U_\bullet h_{t-1} + b_\bullet), \quad g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g)$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \quad h_t = o_t \odot \tanh(c_t)$$

- Notation
  - $W_\bullet$ weights the input and $U_\bullet$ the previous state here, reversing the RNN letters
  - f_t: forget gate, i_t: input gate, o_t: output gate; each has its own weights $W_\bullet$, $U_\bullet$, $b_\bullet$, $\bullet \in \{f, i, o\}$
  - σ: sigmoid; a gate near 0 blocks, near 1 lets through
  - g_t: candidate new content; c_t: cell state, the long-term memory
  - h_t: output; $\odot$: element-by-element product
- GRU (gated recurrent unit): a lighter variant with two gates

Vanishing Gradients: a Bound and a VAR(1) (1/2)

- Training needs the effect of old states on today's state
- Backpropagation through time multiplies one Jacobian per step

$$\frac{\partial h_T}{\partial h_{T-k}} = \prod_{j=0}^{k-1} D_{T-j}\, W, \qquad D_t = \operatorname{diag}\left(\varphi'(W h_{t-1} + U x_t + b)\right)$$

  - $\partial h_T / \partial h_{T-k}$: the matrix of changes in h_T per unit change in the state k steps earlier
  - D_t: diagonal matrix of the slopes φ' of the activation at step t
- Bound, by submultiplicativity $\|AB\|_2 \le \|A\|_2 \|B\|_2$ (Pascanu, Mikolov & Bengio (2013))

$$\left\| \frac{\partial h_T}{\partial h_{T-k}} \right\|_2 \le \left(\gamma_\varphi \|W\|_2\right)^k, \qquad \gamma_\varphi = \sup_z |\varphi'(z)|$$

  - $\|W\|_2$: spectral norm, the largest factor by which W stretches a vector
  - γ_φ: the largest slope of the activation ($= 1$ for tanh)
  - $\gamma_\varphi \|W\|_2 < 1$: the gradient shrinks geometrically with k, i.e. it vanishes
- Scalar case: with $|w| = 0.9$ the gradient falls to 0.12 after 20 steps and to 0.005 after 50 steps (Bengio, Simard & Frasconi (1994))
  - Remedies: clipping the gradient norm (explosion), gating as in the LSTM (vanishing)

Vanishing Gradients: a Bound and a VAR(1) (2/2)

- Linearise around a fixed point h^* of the network

$$h_t - h^* \approx A\,(h_{t-1} - h^*) + DUx_t, \qquad A = DW$$

  - Fixed point: a state that reproduces itself when the input is zero
  - D: the slopes of the activation at h^*
  - The hidden state follows a VAR(1) (vector autoregression) with coefficient matrix A
- $\partial h_T / \partial h_{T-k} = A^k$ is the impulse response of this VAR at horizon k
  - $\rho(A)$: spectral radius, the largest absolute eigenvalue of A
  - $\rho(A) < 1$: stable VAR, gradients decay like $\rho(A)^k$
  - $\rho(A) > 1$: they explode
- A stable RNN near its fixed point is a short-memory model, like any stationary VAR
  - Seminar 14, exercise A4
  - Details in the ▸ Appendix

Why the LSTM Remembers

- Along the cell state the only multiplier is the forget gate
  - $\partial c_T / \partial c_{T-k} = \prod_{j=0}^{k-1} f_{T-j}$ on the direct path
  - $c_t = f_t \odot c_{t-1} + i_t \odot g_t$: an AR(1) with a random coefficient
- Constant forget gate $f = \sigma(b_f)$: an AR(1) with coefficient f
  - b_f: its bias
  - So $\mathrm{Corr}(c_t, c_{t-k}) = f^k$ decays geometrically
  - With $b_f = 0$, the forget gate is $f = 0.5$, and after 50 steps $0.5^{50} \approx 9 \times 10^{-16}$ of the memory is left
  - $b_f = 3$: $f = 0.953$, and $0.953^{50} \approx 0.09$ of the memory is left after 50 steps
  - Half-life 14 steps (the k with $f^k = \frac{1}{2}$, $k = \ln 0.5 / \ln f$)
- Long memory decays hyperbolically: $\rho_k \sim C k^{2d-1}$
  - With ρ_k the autocorrelation at lag k
  - $d \in (0, \frac{1}{2})$ the memory parameter, $C > 0$ a constant
  - S&P 500 absolute returns: the fitted geometric decay gives $\rho_{150} = 0.027$, against 0.059 observed (Appendix)



Sepp Hochreiter, co-inventor of the LSTM

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Appendix: the gradient, measured

Attention: Queries, Keys and Values (1/2)

- **Question of the section**: how can a network link today directly to a day 500 days back?
  - Without the product of Jacobians that makes the RNN gradient vanish
- Each time step asks a question (query) and compares it with the key of every step it is allowed to see

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \qquad Q = XW_Q,\ K = XW_K,\ V = XW_V$$

  - The answer is a weighted average of their values (Vaswani et al. (2017))
  - X: the inputs, one row per time step
  - W_Q, W_K, W_V: learned weight matrices
  - Q, K, V: queries, keys and values

Attention: Queries, Keys and Values (2/2)

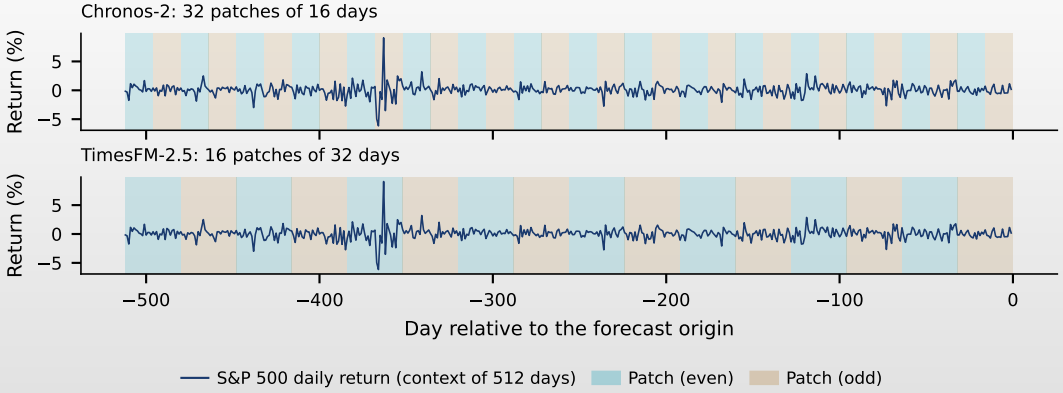
- $QK^\top$: the similarity score of every pair of steps
- d_k: length of a key vector
  - Dividing by $\sqrt{d_k}$ keeps the scores from saturating the softmax
- softmax turns each row of scores $s_1, \ldots, s_n$ into positive weights that sum to one

$$\text{softmax}(s)_j = \frac{e^{s_j}}{\sum_{i=1}^{n} e^{s_i}}$$

- No product of Jacobians over time: any past step is one hop away
- Cost $O(n^2)$: the work grows with the square of the number of inputs n

Appendix: attention in numbers

Patching a 512-Day Context



MFM_ch14_tokenization

Positions and Patches

- **Positional encoding**: attention treats its inputs as a set, so order is added
  - Sinusoids in the original Transformer; rotary embeddings in Chronos-2
  - Rotary embeddings (RoPE): each query and key is rotated by an angle proportional to its position
  - So the score of two steps depends on how far apart they are, not on where they sit in the context
- **Patching** (Nie et al. (2023)): a token is a block of consecutive observations
  - A learned map turns it into a vector of fixed length (embedding)
  - 512 daily returns in patches of 16 days: 32 tokens, 256 times fewer attention scores (32^2 instead of 512^2)
- Chart: the last 512 daily returns of the S&P 500
  - Split as Chronos-2 (16 days) and TimesFM-2.5 (32 days) read them
  - Each shaded block becomes one token
- Global models learn across many series: DeepAR (Salinas et al. (2020)), N-BEATS (Oreshkin et al. (2020)), PatchTST (Nie et al. (2023))
  - Foundation models push this idea to very large corpora

MFM_ch14_tokenization

What Is a Foundation Model?

- **Question of the section**: can a model pretrained on other series forecast a financial series it never saw in training?
- Foundation model: a large network pretrained once on a very broad corpus, then used on new tasks
  - An idea borrowed from large language models (LLM)
- Appeal for finance: one model for thousands of series, no estimation per series
- Three families of models
  - **Chronos** (Ansari et al. (2024)): values become the tokens of a language model
  - **TimesFM** (Das et al. (2024)): a decoder-only Transformer that reads the series in patches
  - **Moirai** (Woo et al. (2024)): any number of variables and a mixture-distribution output (▸ Appendix)
  - Moirai is not tested here because of its non-commercial licence

Zero-Shot Forecasting: What Goes In and What Comes Out

- **Zero-shot** forecasting: the parameters $\hat{\theta}$ are fixed in pretraining

$$\left(\hat{q}_{u_1, T+h}, \ldots, \hat{q}_{u_K, T+h}\right)_{h=1,\ldots,H} = F_{\hat{\theta}}\left(y_{T-C+1}, \ldots, y_T\right)$$

  - At time T the model receives only the recent history of the series (the **context**)
  - $F_{\hat{\theta}}$: the pretrained network, the same for every series
  - C: context length (512 days here); H: forecast horizon
  - $\hat{q}_{u_k, T+h}$: the u_k-quantile of the value at $T+h$, so the output is a quantile grid, not only a point
- Fine-tuning is possible but not used
  - Fine-tuning: further training on the target series
  - We test the "no estimation" promise itself
- The specific risk: the target series may be part of the pretraining corpus (slide on leakage)

Chronos: Time Series as a Language

- Ansari et al. (2024): turn values into tokens and train a language model with cross-entropy
  - The language model: T5, Text-to-Text Transfer Transformer
- Why tokens: the language-model architecture stays unchanged
  - A distribution over tokens imposes no parametric shape (it can be skewed or multimodal)
- Mean scaling and quantisation

$$\tilde{x}_t = \frac{x_t}{s}, \quad s = \frac{1}{C}\sum_{i=\mathbf{1}}^{C} |x_i|, \qquad z_t = \arg\min_j |\tilde{x}_t - c_j|, \quad c_j = -15 + \frac{30\,(j-1)}{4{,}092}$$

  - s: mean absolute value of the C values in the context
  - c_j, $j = 1, \ldots, 4{,}093$: equally spaced bin centres in $[-15, 15]$, $30/4{,}092 = 0.0073$ apart, plus special tokens
  - z_t: the token of x_t, the index of the nearest centre
  - Forecast: sample future tokens z, map them back to values $s\,c_z$, read quantiles from the samples
- Pretraining corpus: public datasets (energy, transport, retail, weather, web, finance and economics)
  - Plus synthetic TSMixup series: random convex combinations of real series
  - And KernelSynth series: paths of Gaussian processes with randomly combined covariance kernels
- Chronos (T5): 8 to 710 million parameters
  - Chronos-Bolt, tested here: 9 to 205 million

The Three Models We Test

Model	Parameters	Patch	Quantile levels	Weights released
Chronos-Bolt (small)	48 M	16	10%–90%	25.11.2024
TimesFM-2.5	231 M	32	mean, 10%–90%	01.10.2025
Chronos-2	119 M	16	21: 1%, 5%, 10%, . . . , 95%, 99%	30.10.2025

- **Chronos-Bolt**: patches and a direct quantile output instead of token sampling
- **Chronos-2** (Ansari et al. (2025)): robust scaling, attention across time and across groups of related series
- **TimesFM** (Das et al. (2024)): pretrained on about 10^{11} time points
- Parameters are counted on the loaded models
  - TimesFM-2.5 is released as timesfm-2.5-200m
- All run on a CPU with a 512-day context
  - The release dates of the weights bound what the models could have seen
  - Other models and their corpora in the Appendix

From a Quantile Grid to VaR and ES (1/2)

- The model returns quantiles q_u at levels $u_1 < \cdots < u_K$: $P(Y \leq q_u) = u$
  - Y: the future value; K: the number of levels
- Why a grid: the same quantiles give the point forecast, the volatility and the tail risk
  - So one model serves all three tests
- Point forecasts from the grid
  - the median $q_{0.5}$, or the mean $\int_0^1 q_u \, du$, the average of all quantiles
  - for any function g with a finite mean: $E[g(Y)] = \int_0^1 g(q_u) \, du$, because Y has the same distribution as q_U with U uniform (e.g. $g = \exp$ when Y is a log variance)
  - If g is increasing, $g(q_u)$ is also the u-quantile of $g(Y)$
  - Check: Y uniform on $(-1, 1)$, $q_u = 2u - 1$, $g(y) = y^2$: $\int_0^1 (2u-1)^2 \, du = 1/3 = E[Y^2]$, although $(2u-1)^2$ is not a quantile of Y^2 (it falls, then rises in u)

From a Quantile Grid to VaR and ES (2/2)

- Tail risk at level α (Chapter 7), for a return Y

$$\mathrm{VaR}_\alpha = -q_\alpha, \qquad \mathrm{ES}_\alpha = -\frac{1}{\alpha}\int_0^\alpha q_u \, du$$

- Below the lowest level the grid is silent
  - Holding q_u at $q_{0.01}$ understates ES
  - Interpolation can err either way (Seminar 14, exercises A1 and A7)
- Chronos-Bolt and TimesFM stop at 10%: a "VaR 1%" request returns the 10% quantile
- The chapter's rule: linear interpolation between levels
  - q_u held flat below the lowest level
  - The worked example is in Further Topics

Pretraining Leakage and Contamination

- A test is out of sample only if the test period was never seen during training
- It can be "seen" in four ways
  - (i) the same series is in the pretraining corpus
  - (ii) correlated series carry the same shocks
  - (iii) developers select model versions on overlapping public benchmarks
  - (iv) our own look-ahead: scalings, windows, seeds or contexts chosen with test data
- The Chronos corpus includes financial series (e.g. daily exchange rates 1990–2016)
  - So we cannot rule out that our series were seen in pretraining
- Remedy for (i)–(iii): a **post-release window** from 3 November 2025
  - A common cutoff after all three sets of weights were published
  - Clean but short: about 220 trading days (320 calendar days for Bitcoin), hence low power, especially for VaR 1%
- (iv) is under our control: every scaling, window and estimate uses only past data
  - Formal protocol in the Appendix

Design of the Return Test

- **Question**: does any model forecast tomorrow's daily return better than the zero forecast?
- Daily returns: S&P 500 and BET from 04.01.2016, Bitcoin from 01.01.2019
  - BET: Bucharest Stock Exchange index
  - 2 693 days (S&P 500), 2 675 days (BET) and 2 818 days (Bitcoin)
  - The test period, about ten years (eight for Bitcoin), includes the March 2020 crash and 2022
- Why three markets
  - S&P 500: the most liquid equity market, where a simple signal would already be traded away
  - BET: a smaller, less liquid market, where stale prices can create predictability
  - Bitcoin: traded around the clock, with a daily standard deviation almost three times larger (3.21% against 1.13% for the S&P 500)
- Models: historical mean, AR(1), LSTM, Chronos-2, Chronos-Bolt and TimesFM-2.5
  - AR(1): $\hat{r}_t = \hat{a} + \hat{b}\, r_{t-\mathbf{1}}$, least squares on the last 1,000 days, re-estimated daily
  - Foundation models: the grid median, a point forecast that extreme values do not move
  - Grid means are reported separately
- Every forecast uses only data up to the previous close

Judging the Return Forecasts (1/2)

- Benchmark: the zero forecast; out-of-sample

$$R^2 = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t r_t^2}$$

  - r_t: realised return; $\hat{r}_t$: forecast from data up to $t-1$
  - $R^2 > 0$: smaller squared errors than the zero forecast
- Why zero rather than the historical mean: the estimated mean adds estimation noise
  - Zero is the benchmark of Gu, Kelly & Xiu (2020), the historical mean that of Campbell & Thompson (2008)
- Grid means ($\int_0^1 q_u \, du$) do worse still
  - On the S&P 500, R^2 is -0.77% (Chronos-2) and -3.42% (TimesFM-2.5)

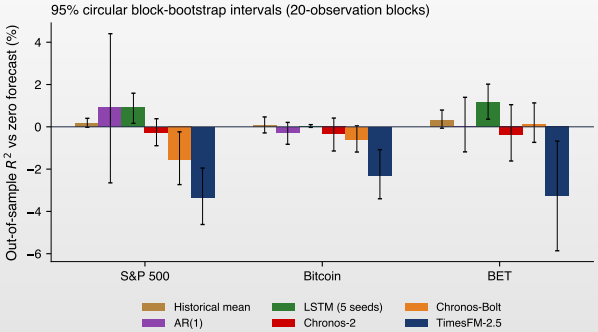
Judging the Return Forecasts (2/2)

- Block-bootstrap confidence interval (CI) for R^2
  - Rebuild a test period of the same length from blocks of 20 consecutive days drawn with replacement, 1,000 times
  - Keep the 2.5% and 97.5% percentiles
  - The blocks keep volatility clusters together
- Diebold–Mariano (DM, Diebold & Mariano (1995)) test on squared errors

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{V}_{NW}/n}}, \qquad d_t = (r_t - \hat{r}_t)^2 - r_t^2$$

  - $\bar{d}$: mean of d_t over the n forecasts
  - $\mathrm{DM} > 0$: the model has larger errors than the zero forecast
  - $\hat{V}_{NW}$: long-run variance of d_t, estimated by Newey–West, a HAC (heteroskedasticity and autocorrelation consistent) estimator
  - Under H_0: $E[d_t] = 0$, DM is approximately standard Normal

Out-of-Sample R^2 for Returns



- R^2 against the zero forecast (%), 95% block-bootstrap CIs
  - Full table with DM p-values in the ▸ Appendix
- S&P 500 and Bitcoin: no model beats zero in the two-sided DM test
- On the S&P 500, the historical mean passes the Clark–West test for nested models ($p = 0.007$)
  - But not the joint SPA test ($p = 0.130$)
  - Both are explained below
- TimesFM-2.5 is significantly **worse** than zero
  - S&P 500: R^2 −3.35%, DM $p = 0.001$
  - Forecast s.d. 0.141%, three times the LSTM's
- BET: LSTM R^2 +1.15% (DM p two-sided 0.014, one-sided 0.007)
  - A small signal, tested below for nesting, multiplicity and stale prices

MFM_ch14_returns

Nested Models: Clark–West against DM

- **The problem**: zero is nested in the historical mean and AR(1)
  - Under H_0 the larger model only adds estimation noise, so DM rejects too rarely
- Clark–West (Clark & West (2007)): add back the noise term and test its mean, one-sided HAC t-test

$$\tilde{d}_t = e_{\mathbf{0},t}^2 - \left[e_{\mathbf{1},t}^2 - (\hat{r}_{\mathbf{0},t} - \hat{r}_{\mathbf{1},t})^2\right]$$

  - $\hat{r}_{\mathbf{0},t}$: zero forecast (small model); $\hat{r}_{\mathbf{1},t}$: forecast of the large model; $e_{\mathbf{0},t}$, $e_{\mathbf{1},t}$: their errors $r_t - \hat{r}_{\cdot,t}$
  - $(\hat{r}_{\mathbf{0},t} - \hat{r}_{\mathbf{1},t})^2$: the cost of estimating coefficients that are zero under H_0
  - H_0: $E[\tilde{d}_t] = 0$; a large positive t: the large model adds information
- One-sided p-values, DM against Clark–West
  - historical mean, S&P 500: DM 0.048, Clark–West 0.007
  - historical mean, BET: DM 0.041, Clark–West 0.003
  - AR(1), S&P 500: DM 0.320, Clark–West 0.055 (the BET AR(1) and rolling windows: Appendix)
- Reading: the correction makes the historical mean significant in both markets
  - But the gain stays small (R^2 of 0.15% on the S&P 500 and 0.30% on the BET)

Many Tests: Bonferroni and Holm

- 6 models $\times$ 3 markets $=$ 18 tests
  - The smallest one-sided DM p is the BET LSTM, 0.007
- The problem: at least one $p < 0.05$ would appear with probability $1 - 0.95^{18} = 0.60$
  - If no model had a signal and the tests were independent
- Bonferroni: with M tests, reject only if $p < 0.05/M$ (here 0.0028)
- Holm: the same idea, applied step by step from the smallest p
- Both control the probability of any false rejection
- Holm-adjusted p over the 18 tests: 0.130, not significant

Many Tests: Romano–Wolf and SPA

- Within the six BET models, tests that use the correlation between the models
  - Romano–Wolf (Romano & Wolf (2005)): step-down adjusted p of the BET LSTM 0.016
  - SPA test (superior predictive ability, Hansen (2005)), H_0: no model beats zero
  - SPA p-values: BET 0.018, S&P 500 0.130, Bitcoin 0.354
- "Some model beats zero on the BET" holds if the BET was chosen in advance as the test family
- Holm-corrected across the three markets, the SPA p-value of the BET becomes $3 \times 0.0185 = 0.056$
  - Just above 5%
- "One particular comparison out of 18 is real" does not hold
  - Fix the test family before looking at the results
- How Holm, Romano–Wolf and SPA work, with a three-test example, in the Appendix

Test 2: the Question and the Choices

- **Question**: do zero-shot foundation models forecast the daily variance of the S&P 500 better than log-HAR?
  - Log-HAR: the standard econometric benchmark
- Why SPY: the S&P 500 ETF is very liquid and has 5-minute prices
  - The realised variance is computed from them
- Why log-HAR: it approximates long memory with three averages and only four coefficients (Corsi (2009))
- Why QLIKE: it ranks forecasts correctly (Patton (2011))
  - Even when the true variance is replaced by a noisy proxy
- Horizon: one day, the same as for the VaR of Test 3

Realised Variance, HAR and the QLIKE Loss (1/2)

- **Realised variance** (RV, Andersen et al. (2003)): sum of squared 5-minute log returns within the trading day

$$\mathrm{RV}_t = \sum_{j=1}^{78} r_{t,j}^2$$

  - $r_{t,j}$: log return of the j-th 5-minute interval of day t
  - 78 intervals in the regular SPY session (6.5 hours); average volatility 11.9% p.a.
- **HAR** (Corsi (2009)): today's RV regressed on yesterday's, the last week's and the last month's averages

$$\mathrm{RV}_t = \beta_0 + \beta_d \mathrm{RV}_{t-1} + \beta_w \overline{\mathrm{RV}}_{t-5:t-1} + \beta_m \overline{\mathrm{RV}}_{t-22:t-1} + \varepsilon_t$$

  - $\overline{\mathrm{RV}}_{t-5:t-1}$, $\overline{\mathrm{RV}}_{t-22:t-1}$: averages over the last 5 and 22 trading days
  - β_d, β_w, β_m: daily, weekly and monthly weights

Realised Variance, HAR and the QLIKE Loss (2/2)

- log-HAR: the same regression on $\log \mathrm{RV}$; forecast $\exp(\hat{y} + \hat{s}^2/2)$
  - $\hat{y}$: fitted log RV; $\hat{s}^2$: variance of the log residuals
  - The factor $e^{\hat{s}^2/2}$ turns a log forecast into the mean of RV if the log error is conditionally Normal
- **QLIKE** (quasi-likelihood) loss (Patton (2011)) of a variance forecast $\hat{h}_t$

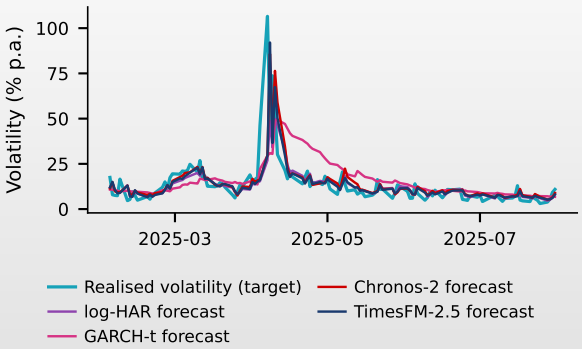
$$L = \frac{\mathrm{RV}_t}{\hat{h}_t} - \log \frac{\mathrm{RV}_t}{\hat{h}_t} - 1$$

  - $L \geq 0$, and $L = 0$ when $\hat{h}_t = \mathrm{RV}_t$
  - Under-forecasts cost more than over-forecasts of the same size
  - Minimised by the conditional mean
  - Ranks forecasts as the true variance σ_t^2 would when RV is a conditionally unbiased proxy, $E[\mathrm{RV}_t \mid \mathcal{F}_{t-1}] = \sigma_t^2$, with $\mathcal{F}_{t-1}$ the information up to $t-1$ (so does MSE)

Design of the Volatility Test

- SPY realised variance, one-day-ahead forecasts from 11.10.2022 to 18.09.2026 (979 days)
  - HAR and log-HAR on a rolling 500-day window
  - Random walk: $\hat{h}_t = \mathrm{RV}_{t-1}$
  - GARCH-t (GARCH(1,1) with Student-t innovations) on daily SPY returns, rescaled to RV units by the ratio of average RV to average squared return over 500 days
  - LSTM on the last 22 values of log RV, retrained every 125 days on 500 days (hidden size 8, three seeds)
- Foundation models on $\log \mathrm{RV}$ (context 512): forecast $\hat{h}_t = \int_0^1 \exp(q_u)\, du$ on the quantile grid
  - q_u are quantiles of log RV, $\exp(q_u)$ those of RV, and their average is the mean of RV
  - On the grid: trapezoid rule over the returned levels, with q_u held at its value at the lowest level below it and at the highest level above it
  - Same rule for all three models, without the TimesFM mean output
  - Post-release window: 218 days from 3 November 2025

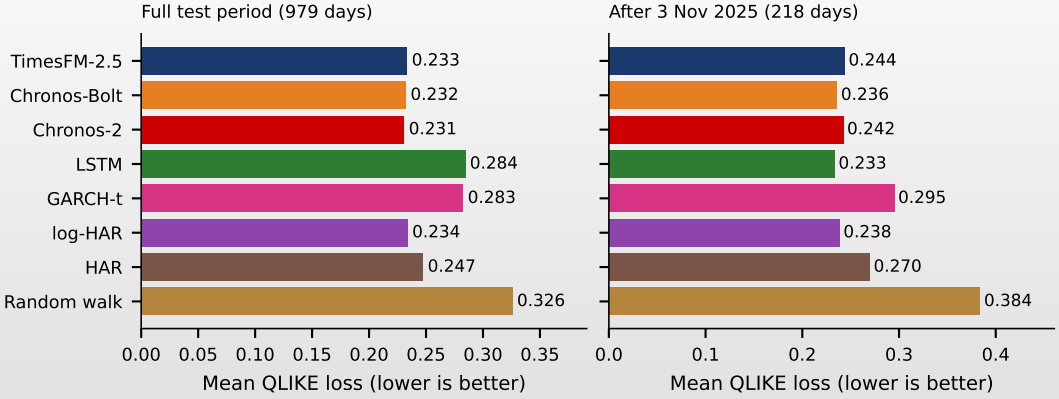
Forecasts Through the April 2025 Shock



- SPY realised volatility (annualised) and one-day-ahead forecasts, February–July 2025
  - Peak 107% p.a. on 07.04.2025, after the US tariff announcements
- No model anticipates the jump
  - All react the next day
  - One-day forecasts extrapolate the past, shocks come from news
- Foundation models and log-HAR decay back at similar speeds
- GARCH-t reacts to daily returns and decays more slowly

MFM_ch14_realised_vol

QLIKE Losses of Eight Volatility Models (1/3)



MFM_ch14_realised_vol

QLIKE Losses of Eight Volatility Models (2/3)

- Bars are average QLIKE losses
  - Left: full test period (979 days)
  - Right: after 3 November 2025 (218 days)
- Chronos-2 (0.231), Chronos-Bolt, TimesFM-2.5 and log-HAR (0.234): DM tests not significant
  - Chronos-2 $p = 0.537$
- The 90% model confidence set (MCS, Hansen, Lunde & Nason (2011)) drops the worst model one at a time
  - Until equal average losses are no longer rejected
- The MCS keeps six of the eight models: only the random walk and GARCH-t are eliminated
- HAR in levels over-forecasts (mean forecast / mean RV = 1.36)
- After release the small LSTM is best (0.233): rankings are unstable
- Full table with DM and MCS p-values in the Appendix

MFM_ch14_realised_vol

QLIKE Losses of Eight Volatility Models (3/3)

- Giacomini–White test (Giacomini & White (2006)): can the loss differential d_t be predicted from yesterday's information?

$$\mathrm{GW} = n\,\bar{Z}^{\top}\hat{\Omega}^{-1}\bar{Z} \;\to\; \chi^2_2, \qquad Z_t = (1,\; d_{t-1})^{\top} d_t$$

  - d_t: QLIKE loss of the model minus that of log-HAR
  - Instruments 1 and d_{t-1}, known at $t-1$
  - H_0: $E[d_t] = 0$ and $E[d_{t-1}d_t] = 0$ jointly
  - $\bar{Z}$: mean of Z_t; $\hat{\Omega} = n^{-1}\sum_t Z_t Z_t^{\top}$; n: number of forecasts
- Result: $p = 0.760$ (Chronos-2) and 0.935 (TimesFM-2.5): no conditional difference
- No difference detected is not proof of equality (power is discussed in the AI section)

Median or Mean under QLIKE

Chronos-2 and TimesFM-2.5 on log RV	QLIKE	Mean forecast / mean RV
Chronos-2, $\int_0^1 \exp(q_u)\,du$	0.231	1.02
Chronos-2, $\exp(q_{0.5})$	0.259	0.79
TimesFM-2.5, $\int_0^1 \exp(q_u)\,du$	0.233	0.94
TimesFM-2.5, $\exp(q_{0.5})$	0.259	0.79

- A theorem (Gneiting (2011)): QLIKE is consistent for the mean
  - The forecast h that minimises its expected value given the information $\mathcal{F}$ is the conditional mean, $\arg\min_h E[L(\mathrm{RV}, h) \mid \mathcal{F}] = E[\mathrm{RV} \mid \mathcal{F}]$
  - Monotone maps preserve quantiles: exp(median of $\log \mathrm{RV}$) is the median of RV, elicited by absolute error, not by QLIKE
  - Lognormal RV, s^2 the variance of log RV: median/mean $= e^{-s^2/2}$
  - The median version is significantly worse (DM +3.60 for Chronos-2) and worse than log-HAR (0.234)

Design of the Tail-Risk Test (1/2): Question and Targets

- **Question**: do foundation models give better VaR 1% and ES 2.5% forecasts than FHS?
  - Either directly or as volatility filters
- Targets: VaR_α and ES_α, computed from the quantiles $q_u(r_{t+1} \mid \mathcal{F}_t)$
  - Definitions: slide From a Quantile Grid to VaR and ES
  - $q_u(r_{t+1} \mid \mathcal{F}_t)$: the u-quantile of tomorrow's return given today's information $\mathcal{F}_t$
  - VaR 1%: the level of regulatory Basel backtesting (coverage tests)
  - The pair (VaR 2.5%, ES 2.5%): the ES level of the FRTB standard (Fundamental Review of the Trading Book)
- Classical models: historical simulation (HS), GARCH(1,1)-t, filtered historical simulation (FHS, McNeil & Frey (2000))
  - 1,000-day window, re-estimated every 20 days
- Why FHS as the benchmark
  - It combines a conditional GARCH volatility with the empirical shape of the tails
  - Without assuming a distribution

Design of the Tail-Risk Test (2/2): the Foundation Models

- **Raw**: Chronos-2 quantiles of tomorrow's return (context: 512 returns)
- **Hybrid**: scale $\hat{\sigma}_t$ = forecast of $|r_t|$ (Chronos-2: grid mean; TimesFM: median)
  - The foundation model reads the series $|r|$: context = the 512 absolute returns before day t
  - Then FHS on $r_t/\hat{\sigma}_t$: the empirical quantile of the standardised returns, times $\hat{\sigma}_t$
- Median or mean matters little here, unlike in Test 2: FHS is scale-free

$$c\hat{\sigma}_t \times q_\alpha\big(r/(c\hat{\sigma})\big) = \hat{\sigma}_t \times q_\alpha\big(r/\hat{\sigma}\big)$$

- A constant factor in $\hat{\sigma}$ (such as a stable median/mean ratio) cancels
  - Only a time-varying factor changes the VaR

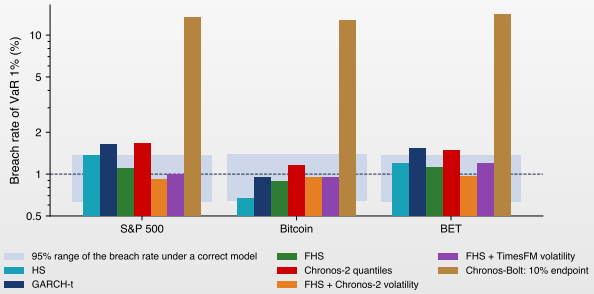
Scoring (VaR, ES) Pairs: the FZ0 Loss

- A functional is elicitable if some loss is minimised, in expectation, by its true value
  - ES alone is not, the pair (VaR, ES) is (Fissler & Ziegel (2016))
- FZ0 loss of a forecast pair, with the daily loss $L_t = -r_t$

$$\mathrm{FZ0} = \frac{1\{L_t > v\}(L_t - v)}{\alpha\, e} + \frac{v}{e} + \log e - 1$$

  - FZ0: the zero-homogeneous Fissler–Ziegel loss, Patton, Ziegel & Chen (2019)
  - v: VaR forecast; $e > 0$: ES forecast; α: tail level; $1\{L_t > v\}$: 1 on a breach day
  - Lower is better: the first term penalises breaches, $\log e$ penalises wide forecasts
- Models compared by DM tests on FZ0 differences and by the MCS of Test 2

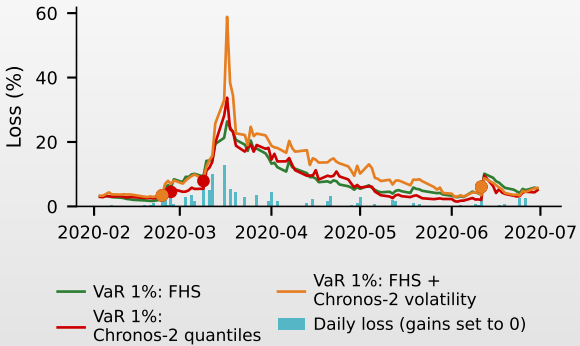
VaR 1% Breach Rates



- Share of days with a loss above VaR 1% (log scale)
  - Light-blue band: 95% range if the model were correct
  - Kupiec (Chapter 8): likelihood-ratio (LR) test that the breach rate equals α, χ^2_1
- Raw Chronos-2: too many breaches on the S&P 500 (1.67%, Kupiec $p = 0.001$) and the BET (1.50%)
- Chronos-Bolt, lowest quantile 10% used as "VaR 1%": 13.5%
- Hybrids: breach rates close to 1% in all three markets
  - Average coverage only
  - Full table in the Appendix

MFM_ch14_tail_risk

The COVID-19 Crash: VaR 1% of Three Models



- S&P 500, February–June 2020 (104 days): daily losses and VaR 1% forecasts
  - Dots: breaches
- Breaches: FHS 2, raw Chronos-2 4, FHS with Chronos-2 volatility 2, HS 12
  - All models are hit in the first days
- Chronos-2 reads the context fast, so its quantiles widen quickly
- The hybrid overshoots: VaR 1% of 58.9% on 17.03.2020
  - Against 26.4% for FHS and a largest loss of 12.8% (Appendix)

MFM_ch14_tail_risk

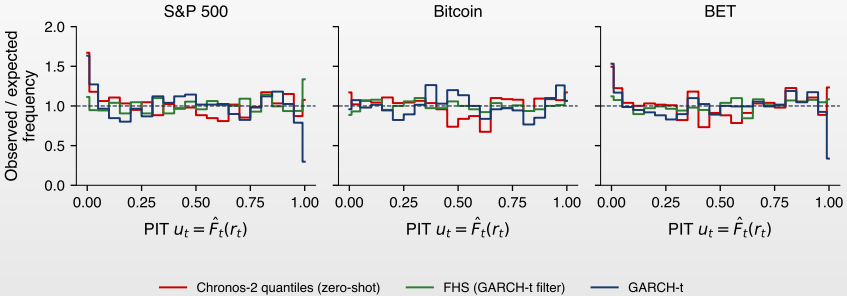
Backtesting Beyond Kupiec: Clustering and the DQ Test

p: indep. / DQ	HS	GARCH-t	FHS	Chronos-2 raw	FHS + Chronos-2	FHS + TimesFM
S&P 500	0.001 / <0.001	0.205 / 0.001	0.045 / 0.021	0.223 / <0.001	0.021 / 0.001	0.029 / <0.001
Bitcoin	0.611 / 0.760	0.261 / 0.463	0.220 / 0.092	0.404 / 0.034	0.026 / 0.004	0.261 / 0.182
BET	0.399 / <0.001	0.258 / 0.082	0.409 / 0.650	0.270 / 0.026	0.475 / <0.001	0.379 / <0.001

- Independence (Christoffersen (1998)): LR test that a breach today does not change the chance of one tomorrow
  - First-order Markov chain, $\sim \chi^2_1$
  - Conditional coverage = Kupiec + independence, χ^2_2
- Dynamic quantile (DQ, Engle & Manganelli (2004)): $\mathrm{Hit}_t = \beta_0 + \sum_{j=1}^{4} \beta_j \mathrm{Hit}_{t-j} + \beta_5 \mathrm{VaR}_t + u_t$
  - With $\mathrm{Hit}_t = I_t - \alpha$ and $I_t = 1$ on a breach day
  - Wald statistic that all six coefficients are zero, $\sim \chi^2_6$
- S&P 500: the hybrids have the right breach rate but fail independence and DQ
  - Every model fails DQ at 5%
- BET: FHS passes both tests, the hybrids fail DQ
  - Right on average is not right conditionally

▸ Appendix: deciles, VaR 10%

The Probability Integral Transform (PIT) and Calibration (1/2)



- PIT $u_t = \hat{F}_t(r_t)$: the forecast probability of a return at or below the one realised (Diebold, Gunther & Tay (1998))
  - If the predictive distribution function $\hat{F}_t$ is correct, the u_t are i.i.d. uniform on $(0, 1)$
- Bars: observed over expected frequency in the 22 intervals of the Chronos-2 grid
  - A bar above 1 means too many outcomes in that region

MFM_ch14_calibration

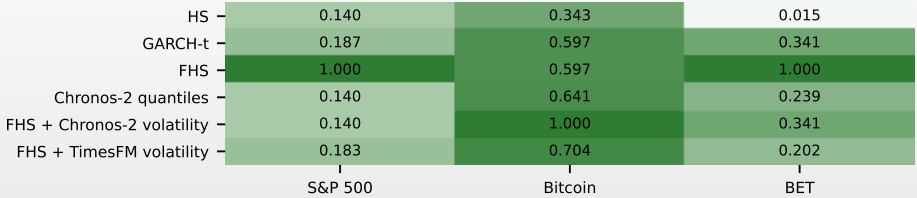
The Probability Integral Transform (PIT) and Calibration (2/2)

- Berkowitz test (Berkowitz (2001)): transform $z_t = \Phi^{-1}(u_t)$, standard Normal if the forecast is correct
  - Test $\mu = 0$, $\sigma = 1$ for z_t, censored outside [1%, 99%]
  - Φ^{-1}: standard Normal quantile function; μ, σ: mean and standard deviation of z_t; $\hat{\sigma} > 1$: tails too thin in the forecast
  - Chronos-2 rejected on the S&P 500 ($p = 0.019$, $\hat{\sigma} = 1.03$) and the BET; FHS never
- Quantile score (QS): the average pinball loss over the K levels u_k of the grid

$$\mathrm{QS} = \frac{2}{K} \sum_{k=1}^{K} \left(1\{r_t < q_{u_k}\} - u_k\right)\left(q_{u_k} - r_t\right)$$

  - A grid analogue of the CRPS (continuous ranked probability score, Gneiting & Raftery (2007))
  - q_{u_k}: forecast quantile at level u_k; lower is better
  - On the S&P 500 and the BET, Chronos-2 is worse than FHS in the centre and in the tails (S&P 500 DM +2.51 and +3.65)
  - On Bitcoin the difference is not significant; table in the Appendix

Comparing (VaR 2.5%, ES 2.5%) Forecasts: the MCS



- p-values of the 90% model confidence set (Hansen, Lunde & Nason (2011)) for the FZ0 loss
  - Darker green means a higher p-value
  - The only model eliminated ($p < 0.10$) is HS on the BET ($p = 0.015$, the near-white cell)
  - FZ0 table in the ▸ Appendix
- S&P 500: FHS has the lowest FZ0
  - The hybrids have the right breach rate but clustered breaches and a larger VaR
  - Average VaR 1% over the 2 693 test days: 3.20% for the Chronos-2 hybrid vs 2.78% for FHS
- Pairwise DM is more powerful than the MCS
  - FHS beats the Chronos-2 hybrid on the S&P 500 (DM +2.65, $p = 0.008$) and the BET (DM +2.36)

MFM_ch14_tail_risk

After the Release: Few Breaches, Little Power

Breaches after 3.11.2025	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (220 days)	0	2	2	2	2	2
Bitcoin (320 days)	3	3	4	4	3	4
BET (217 days)	4	4	4	5	1	2

- Expected breaches: 2.2 (S&P 500), 3.2 (Bitcoin), 2.2 (BET)
  - No contamination is possible, but the tests have little power
  - Exact two-sided Kupiec test at 5% against a true rate of 1.67%: power (probability of rejecting) 0.19 at $T = 220$ days
  - Actual size (rejection rate when the model is right) 0.13; power 0.50 at $T = 1{,}000$; 0.87 at $T = 2{,}693$
  - 80% power first at about 2 130 days (8.4 years), for good from 2 320
  - Against 2%: power 0.29 at $T = 220$ (Seminar 14, exercises A2 and A9)
- Low power is not zero power
  - HS has 0 breaches on the S&P 500, and $x = 0$ is in the Kupiec rejection region at $T = 220$
  - A non-rejection does not establish calibration

Recalibrating Quantiles: Adaptive Conformal Inference

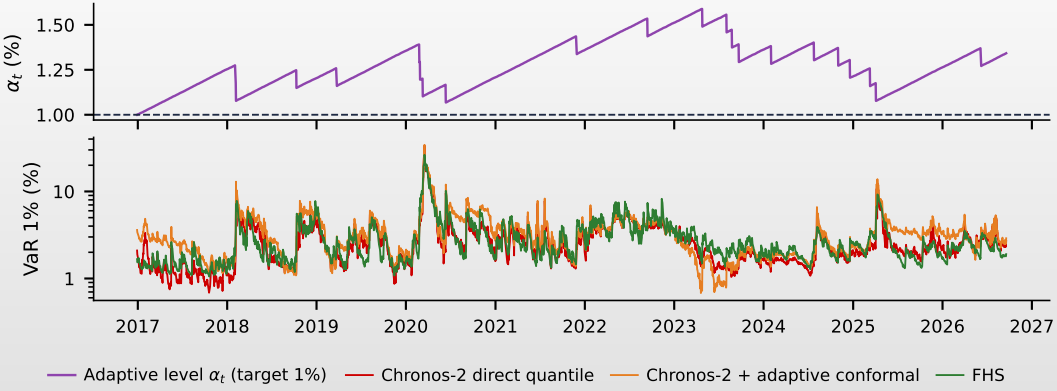
- Idea: shift the model's quantile by the error it made recently
  - And adapt the level after each breach
- Score $E_t = \hat{q}_{\alpha,t} - r_t$
  - Corrected quantile $\hat{q}_{\alpha,t} - \hat{Q}_{\mathbf{1}-\alpha_t}$
  - $\hat{q}_{\alpha,t}$: the model's α-quantile for day t; $E_t > 0$: the return fell below it
  - $\hat{Q}_{\mathbf{1}-\alpha_t}$: the empirical $(1-\alpha_t)$ quantile of the last 250 scores
- Adaptive level of adaptive conformal inference (ACI, Gibbs & Candès (2021))

$$\alpha_{t+\mathbf{1}} = \alpha_t + \gamma(\alpha - \mathrm{err}_t), \qquad \mathrm{err}_t = 1\{r_t < \hat{q}_{\alpha,t} - \hat{Q}_{\mathbf{1}-\alpha_t}\}, \qquad \alpha_{\mathbf{1}} = \alpha$$

  - α: target level (1%); $\mathrm{err}_t = 1$ on a breach
  - $\gamma > 0$: step size, here 0.001; $\gamma = 0$: plain split conformal
  - After a breach α_t falls (the next VaR is wider); after a calm day it rises slowly

▸ Appendix: the ACI guarantee

Adaptive Conformal VaR 1% on the S&P 500 (1/2)



MFM_ch14_calibration

Adaptive Conformal VaR 1% on the S&P 500 (2/2)

- Top: adaptive level α_t (target 1%), mostly above 1%
  - The score shift alone would over-widen the VaR (with $\gamma = 0$, plain split conformal: 16 breaches, average VaR 1% 3.58%)
  - So ACI loosens the level
- Bottom: VaR 1% of raw Chronos-2, Chronos-2 with ACI (adaptive conformal inference), FHS (log scale)
- ACI brings the breaches near 1% (S&P 500: 21 vs 24.4 expected)
  - Conditional coverage is not rejected in any market
- DQ still rejects on the S&P 500 and the BET
  - The VaR is wider (average VaR 1% over the 2 443 days after the 250 calibration days: 3.36% vs 2.85% for FHS on the same days)
  - FZ0 is significantly higher
- The guarantee concerns the average of err_t, not the value of α_t
- Proper scoring rules reward sharpness subject to calibration (Gneiting & Raftery (2007))
  - Calibration is necessary, not sufficient

MFM_ch14_calibration

Case Study: Is Complexity a Virtue in Return Prediction? (1/2)

- **Questions** (Kelly, Malamud & Zhou (2024))
  - must a return-prediction model have far fewer parameters P than observations T?
  - can models with complexity $c = P/T > 1$ do better out of sample?
- Setting (eqs. 1–2): next month's return is an unknown function of today's predictors

$$R_{t+\mathbf{1}} = f(G_t) + \varepsilon_{t+\mathbf{1}}, \qquad f(G_t) \approx \sum_{i=\mathbf{1}}^{P} S_{i,t}\,\beta_i$$

  - Approximated by a wide network
  - $R_{t+\mathbf{1}}$: market excess return; G_t: vector of predictors known at t; $\varepsilon_{t+\mathbf{1}}$: unpredictable part
  - $S_{i,t}$: the i-th non-linear feature of G_t (a hidden unit of a one-hidden-layer network); β_i: its weight
  - Universal approximation (Hornik, Stinchcombe & White (1989))
- **Why it matters**: a theoretical case for the large models of this chapter
  - Against the parsimony rule of classical econometrics

Case Study: Is Complexity a Virtue in Return Prediction? (2/2)

- **Link to Test 1**: no model beats the zero forecast on the S&P 500 and Bitcoin
  - After the multiple-testing correction
  - The paper asks whether a negative out-of-sample R^2 means no economic value
  - Economic value: Sharpe ratio of the market-timing return $\pi_t R_{t+1}$, with position $\pi_t = \hat{\beta}' S_t$ (eq. 6)
  - The forecast itself is the position
  - $\hat{\beta}$: estimated weights; S_t: vector of the P features at t
- **Theory** (Sections III–IV): expected out-of-sample R^2 and timing Sharpe ratio rising with c
  - For a misspecified model with optimally chosen ridge shrinkage, under the assumptions of the paper
  - Near $c = 1$ (the interpolation boundary) least squares fits the training data exactly and its variance explodes
  - Beyond it, performance recovers

Case Study: Data and Method

- **Data** (Section V.A): monthly excess return of the CRSP value-weighted index, 1926–2020
  - CRSP: Center for Research in Security Prices
  - The 15 predictors of Welch & Goyal (2008) (14 variables and the lagged market return)
  - Returns scaled by their trailing 12-month volatility, predictors by an expanding-window standard deviation
  - 36 months of warm-up, so the sample starts in 1930
- **Random Fourier features** (RFF; eq. 20, Rahimi & Recht (2007))

$$S_{i,t} = \big(\sin(\gamma\,\omega_i' G_t),\ \cos(\gamma\,\omega_i' G_t)\big)', \qquad \omega_i \sim N(0, I), \quad \gamma = 2$$

  - Each feature is a sine or cosine of a random combination of the predictors
  - ω_i: random weight vector, drawn once from the standard Normal distribution (I: identity matrix) and then fixed
  - γ: degree of non-linearity, larger γ gives faster oscillating features
  - A wide two-layer network whose first-layer weights are random; only the output weights β are estimated
  - P counts columns: $P/2$ vectors ω_i, each giving a sine and a cosine (so $P = 12\,000$ uses $6\,000$ vectors ω_i)

▸ Appendix: ridge and design

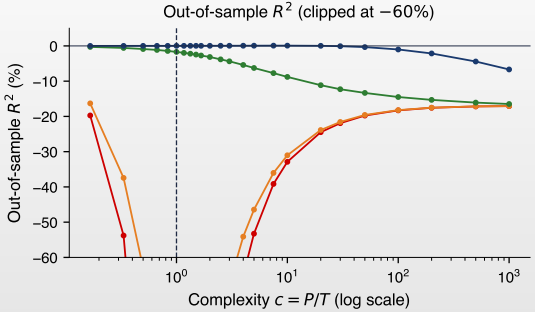
Case Study: Key Results

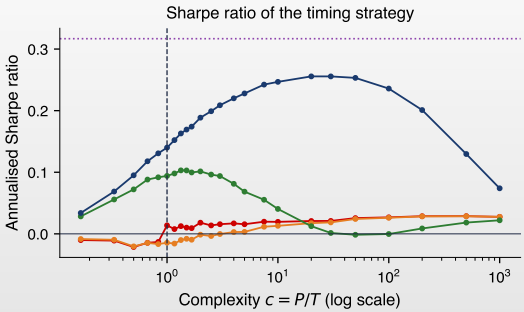
Model ($T = 12$)	R^2 (%)	Sharpe [t]	Inf. ratio [t]	Max loss	Skewness
Linear, $z = 0^+$	< -100	-0.11 [-1.0]	-0.16 [-1.6]	98.5	-0.9
Linear, $z = 10^3$	-3.8	0.46 [4.4]	0.33 [3.1]	2.4	-0.1
RFF, $c = 1\,000$, $z = 10^3$	0.6	0.47 [4.5]	0.31 [2.9]	1.2	2.5

Source: Table I, Panel A, of the paper; information ratio against the market; maximum monthly loss in standard deviations

- Ridgeless kitchen sink ($P = 15 > T = 12$): R^2 below -100%, Sharpe ratio -0.11
  - Reading: at the interpolation boundary forecasts explode, reproducing the negative verdict of Welch & Goyal (2008)
- Same predictors with $z = 10^3$: $R^2 = -3.8\%$, yet Sharpe ratio 0.46 ($t = 4.4$)
  - Reading: R^2 penalises the scale of forecasts; timing profits depend on their correlation with returns (footnote 5)
- RFF, $c = 1\,000$: information ratio 0.26 ($t = 2.5$) against the best linear model
  - With $T = 60$ and 120, 0.25 and 0.24 against the market ($t = 2.3$ and 2.2; Panels B–C)
  - Reading: nonlinearity extracts more from the same predictors, with a smaller maximum loss and positive skewness
- Positions (Figure 10): almost always long
  - Cut before 14 of 15 NBER recessions, out of sample
  - Reading: the model learns the sign restriction of Campbell & Thompson (2008) without being given it

Case Study: Complexity on Course Data (1/2)





$z = 10^{-3}$ $z = 10^{-1}$ $z = 10^{1}$ $z = 10^{3}$ Linear, 7 predictors, $z = 10^3$

MFM_ch14_virtue_complexity

Case Study: Complexity on Course Data (2/2)

- Figures 7–8 of Kelly, Malamud & Zhou (2024) on course data: SPY excess returns, 7 predictors
  - 353 test months (February 1997–August 2026)
  - $T = 12$ complete observations, $\gamma = 2$, 1 000 draws
  - Dashed: $c = 1$; dotted: linear, $z = 10^{3}$
  - Data in the Appendix
- Nearly ridgeless ($z = 10^{-3}$): $R^2 = -1\,834\%$ at $c = 1$, recovering to -17.1% at $c = 1\,000$
  - The double descent of Figure 7 (left panel cut at -60%)
- $z = 10^{3}$: Sharpe ratio 0.14 at $c = 1$, 0.26 at $c = 20$, 0.07 at $c = 1\,000$ ($t = 0.4$)
  - Linear with $z = 10^{3}$: Sharpe 0.32 ($t = 1.7$), not beaten by RFF
  - Information ratio of RFF 0.03 against the linear model, $t = 0.1$; -0.17 against SPY
- Reading: on course data, complexity helps up to $c \approx 20$, not up to $c = 1\,000$

MFM_ch14_virtue_complexity

Case Study: Limits and an Open Question

- **Limits** stated by the authors
  - Theory for linear ridge models on fixed random features and a single asset
  - A panel with cross-asset covariances is left open (Introduction, footnote 2)
  - Predictability spread evenly across features (Assumption 4)
  - Sparse LASSO-type models left to future research (footnote 22)
  - No optimal degree of nonlinearity γ
  - Average timing returns about half as large in 1975–2020 as in 1930–1974 (Section V.F)
- **Course data** (Seminar 14, exercise C3): 7 of the 15 predictors and about 30 years
  - The power of a two-sided 5% test for an information ratio of 0.31 is only about 39% ($t \approx 0.31\sqrt{29.4} \approx 1.7$)
- **Open question**: learned instead of random features
  - The embeddings of a pretrained foundation model such as Chronos-2 as S_t
  - Is complexity still a virtue, on a panel of course markets?

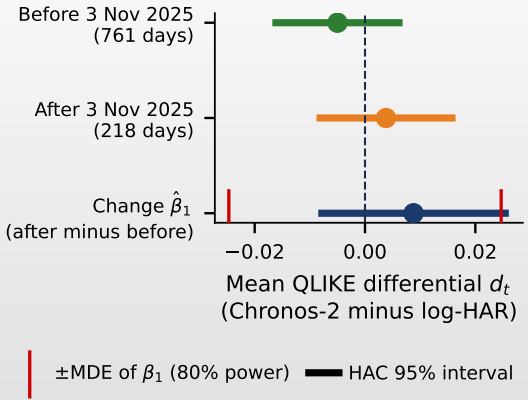
An Open Question: Skill or Memory?

- Test 2: zero-shot Chronos-2 is not distinguishable from log-HAR on SPY realised variance
- Rahimikia, Ni & Wang (2025): on daily excess returns across global markets, off-the-shelf foundation models do poorly
  - Zero-shot and after fine-tuning
  - Models pretrained from scratch on financial data gain
- Lopez-Lira, Tang & Zhu (2025): once a model has seen the realised values, its counterfactual forecasting skill is not identified
  - Detected memorisation is only a lower bound
  - Shown for LLMs; the logic applies to any pretrained forecaster
- Open questions
  - is the parity learned structure (long memory, mean reversion) or recall of the test period?
  - can a short post-release window tell?
- A test: $d_t = L(\hat{h}_t^{FM}, \mathrm{RV}_t) - L(\hat{h}_t^{\log \mathrm{HAR}}, \mathrm{RV}_t)$
  - The QLIKE loss L of the foundation model (FM) minus that of log-HAR
  - Memorisation implies $E[d_t \mid \text{post}] > E[d_t \mid \text{pre}]$: the FM loses its edge after the release of its weights
  - H_0: equality; a rejection may also reflect a regime change

The Discovery Loop in Five Steps

- **Literature**: ask an assistant for evaluations of time-series foundation models on financial data since 2024
  - Check every arXiv identifier and read the corpus documentation yourself
- **Hypotheses**: predictions that separate memorisation from skill
  - Pre vs post release, never-public series, simulated series with the same dynamics
- **Data and code**: from the chapter's volatility forecasts, the QLIKE differential Chronos-2 minus log-HAR
  - A pre/post test with HAC errors and the minimum detectable effect (MDE)
- **Robustness**, fixed in advance: a placebo release date one year earlier must show no break
  - Simulated series cannot be memorised
- **Critique**: ask the assistant to attack "no memorisation, because the post-release test is insignificant"
  - Expect low power and different regimes
  - Reusable prompts in the ▸ Appendix

Mini-Case: a Test Without Power (1/3)



MFM_ch14_ai_discovery

Mini-Case: a Test Without Power (2/3)

- Typical assistant proposal: "the parity of Chronos-2 comes from memorised data, so its edge must vanish after release"
- Chart: mean QLIKE differential d_t (Chronos-2 minus log-HAR) before and after the release
  - Green: before; orange: after
  - With 95% HAC intervals
- Bottom row: the estimated change $\hat{\beta}_1$, with red lines at $\pm$MDE
  - MDE: the smallest effect the test can detect
- Mean d_t: -0.0050 before 3 Nov 2025 (761 days), $+0.0038$ after (218 days)
- Change $+0.0088$, HAC $t = 1.00$, $p = 0.32$

MFM_ch14_ai_discovery

Mini-Case: a Test Without Power (3/3)

- Minimum detectable effect: MDE $= (z_{0.975} + z_{0.8})\,\mathrm{SE}_{HAC}(\hat{\beta}_1) = 0.0247$
  - $z_{0.975} = 1.96$, $z_{0.8} = 0.84$: standard Normal quantiles, 5% two-sided test, 80% power
  - $\hat{\beta}_1$: estimated post-release shift in the mean of d_t; SE_{HAC}: its HAC standard error
  - 4.9 times the pre-release gap
  - Still 0.0169 with an infinite post-release window
- Why a floor: HAC errors give each period its own long-run variance V

$$\mathrm{SE}^2 \approx \frac{V_{pre}}{761} + \frac{V_{post}}{218}$$

  - More post-release days remove only the second term, leaving $2.80\sqrt{V_{pre}/761}$
- Reading: "no memorisation" is not a finding
  - Power must come from the cross-section, not from waiting

MFM_ch14_ai_discovery

▸ Appendix: pitfalls to check

Your Turn: a Project Seed

- Questions (where memorisation is impossible)
  - does zero-shot skill for daily variance survive after release?
  - does it survive on simulated series?
  - does it survive on less-covered markets?
- Replicate first: the chapter's QLIKE of log-HAR (0.234) and Chronos-2 (0.231)
  - DM $p = 0.537$
  - The 218-day post-release window
- Deliverables
  - A pre-registered power calculation
  - A panel of series from the course data with a range-based variance proxy and a proxy-robust loss (Patton (2011))
  - Pre/post tests with placebo release dates; the MCS across models
  - A referee-style critique and your reply
- Log each prompt that shaped a decision in `AI_USE.md`
  - Each invented reference, leak or wrong number caught in `AI_ERRORS.md`, with the test that caught it
  - Project templates

When to Use What

Target	Reference model	Where deep learning can help
Daily index returns	Zero forecast	Less efficient markets (BET); many assets, many features
Realised volatility	log-HAR	Zero-shot forecasts for thousands of series without estimation
VaR 1%, ES 2.5%	FHS (GARCH filter)	Volatility filter in FHS or conformal recalibration

- Foundation models are cheap to try: no estimation, on a CPU
- Their value is breadth (many series, no tuning), not accuracy on a single well-studied series

Key Takeaways

- LSTM gates can mitigate vanishing gradients
  - With a forget-gate bias of 0 they still vanish
- Attention connects distant time steps directly
- Foundation models forecast zero-shot from a context window and return quantiles
- The three tests
  - Returns: after the multiple-testing correction, nothing beats zero on the S&P 500 and Bitcoin
  - Volatility: foundation models not distinguishable from log-HAR
  - Tails: use them only as volatility filters
- Evaluate every new model against a strong benchmark and after the release of its weights
  - With inference that matches the design: Clark–West for nested models
  - SPA or Romano–Wolf for many models
  - PIT, quantile scores and DQ for distributions
- Further reading in the Appendix

Check Yourself, and Next: Chapter 15

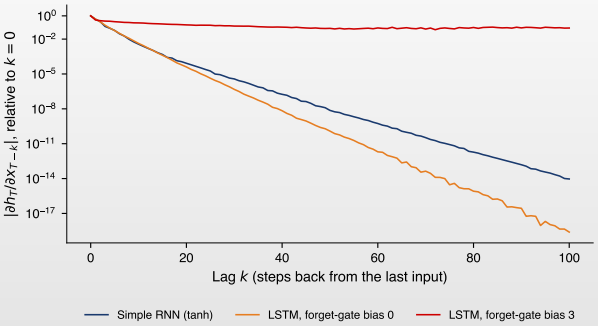
Retrieval questions

- Why does a stable RNN forget at the rate at which a VAR(1) impulse response decays?
- Why is $\exp(q_{0.5})$ of log RV a biased forecast under QLIKE?
- Why can a hybrid VaR pass Kupiec and still lose the FZ0 comparison?
- Why is "not significant after release" not evidence against memorisation?

Next: LLMs and sentiment analysis

- Can text read by a large language model forecast returns or volatility?
- The same threats return: look-ahead, memorisation of the test period, weak benchmarks
- The same tools: post-release windows, Clark–West, multiplicity control

The Vanishing Gradient, Measured



- $|\partial h_T/\partial x_{T-k}|$ by automatic differentiation, relative to $k = 0$ as on the vertical axis
  - 16 hidden units, 50 random initialisations, same Normal inputs
- Simple RNN: 8.3×10^{-5} at lag 20, 7.4×10^{-9} at lag 50
  - Nothing is learned from inputs more than a few weeks back
- LSTM, forget-gate bias 0: 1.2×10^{-10} at lag 50
  - No better than the RNN: $f \approx 0.5$ halves the memory at every step
- LSTM, bias 3: 8.4% at lag 50
  - As predicted by $0.953^{50} \approx 0.09$
  - Initialise the forget-gate bias at a positive value

MFM_ch14_vanishing_gradient

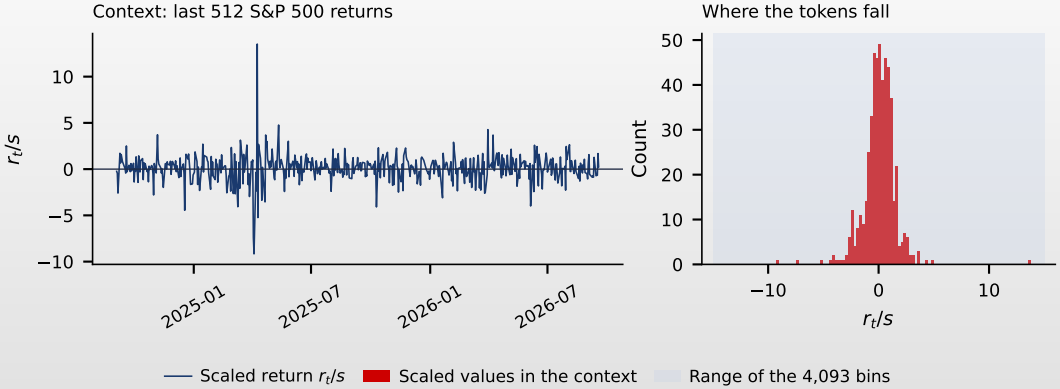
Attention in Numbers: Three Steps and a Causal Mask

- Three time steps, $d_k = 1$: queries $q = (1, 0, 2)$, keys $k = (0.5, 1, -1)$, values $v = (1, 2, 3)$
- Step 3: scores $q_3 k_j = (1, 2, -2)$
  - Softmax weights $e^{s_j} / \sum_i e^{s_i} = (2.718, 7.389, 0.135)/10.243 = (0.265, 0.721, 0.013)$
  - Output $0.265 \cdot 1 + 0.721 \cdot 2 + 0.013 \cdot 3 = 1.75$: step 3 draws mostly on step 2, whose key matches its query best
- **Causal mask**: in a forecaster that predicts every next step, step t may use only steps $j \leq t$

$$\text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V$$

  - Equivalently add $M_{tj} = 0$ for $j \leq t$, $M_{tj} = -\infty$ for $j > t$
  - Step 1 then puts weight 1 on itself (output 1)
  - Without the mask its scores $(0.5, 1, -1)$ give weights $(0.35, 0.57, 0.08)$: 57% from step 2, a later day
  - With the mask, step 2 ($q_2 = 0$, equal scores on steps 1 and 2) outputs $(1 + 2)/2 = 1.5$
- Every value in a 512-day context is already observed at the forecast origin
  - So a model reading only the context may attend in both directions
  - The mask matters where the model is trained to predict later positions from earlier ones

What Chronos Sees: Scaled Returns and Tokens (1/2)



MFM_ch14_tokenization

What Chronos Sees: Scaled Returns and Tokens (2/2)

- Context: the last 512 S&P 500 daily returns
  - Scale $s = 0.672\%$
  - Scaled returns from -9.2 to 13.5, 96% within ± 3
- The 512 returns use 336 distinct tokens out of 4,093
  - Scaling makes the model free of units, so the volatility level must be read from the context
- A crash day with return M (in %) enters the context and the oldest return x_{out} leaves it
  - s rises by $(|M| - |x_{\text{out}}|)/C \approx |M|/C$ percentage points
  - The relative rise, about $|M|/(Cs)$, is large in a short context and small in a 512-day one
  - Seminar 14, exercises A3 and A6

MFM_ch14_tokenization

Forecasting Transformers: the Transformer Block

- **Transformer block** (Vaswani et al. (2017)): multi-head attention, then a small MLP applied to each position separately

$$Z = \mathrm{LN}(X + \mathrm{MHA}(X)), \qquad H = \mathrm{LN}(Z + \mathrm{MLP}(Z))$$

  - Each part has a skip connection and a normalisation
  - $\mathrm{MHA}(X) = [\mathrm{head}_1, \ldots, \mathrm{head}_m]\, W_O$ (multi-head attention): m attentions in parallel, each with its own W_Q, W_K, W_V, concatenated and projected by W_O
  - LN (layer normalisation): each row is standardised, then rescaled by learned parameters
  - $X + \mathrm{MHA}(X)$: the skip connection, which lets the gradient pass straight through the block
  - A model stacks several blocks: H is the input of the next block

Forecasting Transformers: Architectures and Losses

- Three ways to assemble the blocks
  - **Encoder**: attention in both directions inside the context; Chronos-2 is encoder-only and returns the quantile grid in one pass (Ansari et al. (2025))
  - **Decoder-only**: causal mask, trained to predict the next patch at every position: TimesFM (Das et al. (2024))
  - **Encoder–decoder** (T5, used by the original Chronos): the encoder reads the context; the decoder writes the forecast one token at a time
- **Cross-entropy** (original Chronos): predicting the next token is a classification over 4,093 bins

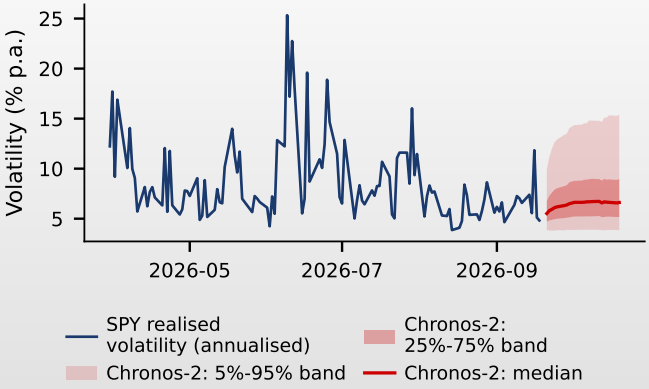
$$\mathcal{L}(\theta) = -\sum_{h=\mathbf{1}}^{H} \log \hat{p}_\theta \left(z_{C+h} \mid z_{\mathbf{1}}, \ldots, z_{C+h-\mathbf{1}} \right)$$

  - $\hat{p}_\theta(z \mid \cdot)$: the probability the model gives to the token that occurred, given the earlier tokens
- **Robust scaling** (Chronos-2, Ansari et al. (2025)): $\tilde{x}_t = \sinh^{-\mathbf{1}} \left((x_t - \hat{\mu})/\hat{\sigma} \right)$
  - $\hat{\mu}$, $\hat{\sigma}$: the mean and standard deviation of the context
  - $\sinh^{-\mathbf{1}}$ grows like a logarithm for large values and so damps outliers

From a Quantile Grid to VaR and ES

- Worked example, S&P 500, day after 18.09.2026: $q_{0.01} = -2.704\%$, $q_{0.05} = -1.400\%$
  - VaR 1% $= 2.704\%$
  - linear interpolation between the two levels: $q_{0.025} = q_{0.01} + \frac{0.015}{0.04}(q_{0.05} - q_{0.01}) = -2.215\%$, so VaR 2.5% $= 2.215\%$
  - ES 2.5%: q_u flat below 1%, trapezoid rule between 1% and 2.5%
- $\mathrm{ES}_{2.5\%} = -\frac{1}{0.025}\left[0.01\, q_{0.01} + 0.015\, \frac{q_{0.01}+q_{0.025}}{2}\right] = 2.56\%$
  - In numbers: $40 \times [0.01\times\ 2.704 + 0.015\times\ 2.459] = 40\times\ 0.06392 = 2.557$, i.e. 2.56%
- Reading: ES 2.5% is close to VaR 1%
  - ES/VaR 2.5% $= 1.15$ is below the Normal value 1.19, although this grid has a heavier left tail than the Normal
  - The flat tail below 1% understates ES (Seminar 14, exercise A7)

A Zero-Shot Fan Chart for Volatility (1/2)



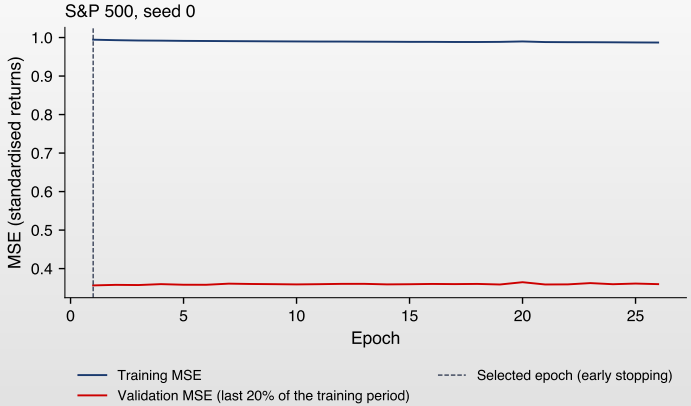
MFM_ch14_fan_chart

A Zero-Shot Fan Chart for Volatility (2/2)

- Chronos-2 on log realised variance of SPY (512-day context)
  - SPY: the exchange-traded fund (ETF) that tracks the S&P 500
  - 22 trading days ahead from 18.09.2026
  - Bands: quantiles 5–95% and 25–75%
- Last volatility 4.8% p.a., 20-day average 6.6%
  - Median forecast at 22 days 6.6%, 90% band 3.8–15.4%
- Mean reversion, as in GARCH and HAR (Chapters 5, 9), with no estimation on SPY
  - The band is skewed upwards partly by construction: volatility is $\sqrt{252\, e^{q}}$ of a log-variance quantile q
  - A convex map stretches the upper side
  - On the log-volatility scale (half the log-variance), the band still lies 0.55 below and 0.85 above the median, so the model itself also leans towards spikes

MFM_ch14_fan_chart

The LSTM for Returns: Training Curves



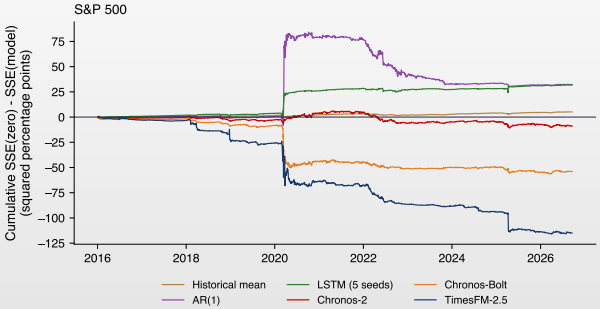
MFM_ch14_returns

The LSTM for Returns: Architecture and Early Stopping

- Input: the last 20 returns, standardised on the training period
  - $h = 16$ hidden units, $d = 1$ input per step, linear output
  - $4(hd + h^2 + 2h) + h + 1 = 1\,233$ parameters (4: the three gates and the candidate; $2h$: two bias vectors per gate)
- MSE, Adam, weight decay 10^{-4}
  - Trained once on up to 4,000 days, the last 20% being the validation block
  - Early stopping: training stops after 25 epochs without a lower validation MSE (patience), and the best weights are kept
  - Forecast = average of 5 seeds
- Chart (S&P 500, seed 0): training and validation MSE by epoch, on standardised returns
  - The dotted vertical line marks the epoch kept
- Training stops 26 epochs in, and the weights of epoch 1 are kept
  - The kept network has seen the training data once and stays close to its random start: one reason why the seed matters
  - Validation MSE is lower than training MSE mainly because the last years of the training period were calmer
- Little to learn: forecast s.d. 0.047% against 1.13% for returns

MFM_ch14_returns

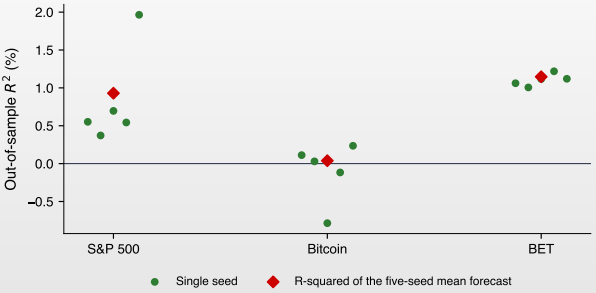
Cumulative Squared-Error Gains on the S&P 500



- Curve SSE(zero) − SSE(model) $= \sum_{s \le t} \left[r_s^2 - (r_s - \hat{r}_s)^2 \right]$
  - SSE: sum of squared errors, in $\%^2$
  - Rises when the model beats the zero forecast, falls when it loses
- All curves stay near zero in calm years and jump in March 2020
  - One month decides the ranking
- AR(1) gains in 2020 from the negative autocorrelation of crash days
  - Then gives part of the gain back
- TimesFM-2.5 loses steadily
- Report the path, not only the final R^2

MFM_ch14_returns

The Seed Lottery



- Out-of-sample R^2 of five LSTMs that differ only in the random seed, and of their average
- Single seeds: variation as large as the effect
  - Bitcoin: from -0.79% to $+0.24\%$
  - S&P 500: from $+0.37\%$ to $+1.96\%$
- BET: robust across seeds
  - All 5 seeds between $+1.01\%$ and $+1.22\%$
- Choosing the best seed after seeing the test is backtest overfitting (Chapter 13)

MFM_ch14_returns

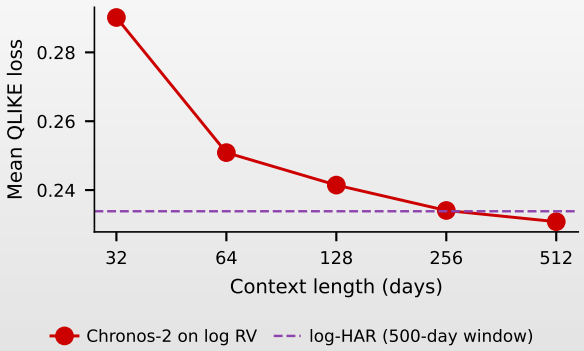
Is BET Predictability Real? Non-Synchronous Trading

- Stale prices of thinly traded constituents make index returns positively autocorrelated (Lo & MacKinlay (1990))
  - Even if every price is efficient
  - First-order autocorrelation since 2016: BET 0.06, S&P 500 −0.14
  - The LSTM forecast has correlation 0.88 with the previous day's BET return: it mostly learned this autocorrelation
- Tradable check: the BET-tracking ETF Patria-TVBETETF, 2 629 common days, same LSTM signal
  - Same day: R^2 2.88% (BET 1.26% on the same 2 629 days)
  - But the ETF is thinly traded too (autocorrelation 0.07, 4.2% zero-return days)
  - Executed one day later: R^2 0.57%, Clark–West $p = 0.003$
- Reading: the evidence is consistent with stale-price effects
  - A small signal survives the execution lag, before transaction costs (Chapter 2)

Computing the Grid Mean: the Tail Rule Matters

- Toy grid: levels 25%, 50%, 75%, log-quantiles $q = -1, 0, 1$; target $E[Y] = \int_0^1 e^{q_u}\, du$
  - Equal weights on the three points: $(e^{-1} + 1 + e)/3 = 1.36$
  - Course rule (trapezoid, flat below 25% and above 75%): $\frac{1}{4}\left[e^{-1} + \frac{e^{-1}+1}{2} + \frac{1+e}{2} + e\right] = 1.41$
  - A Normal log Y with these quartiles ($s = 1/z_{0.75} = 1/0.674 = 1.48$, $z_{0.75}$ the 75% standard Normal quantile) has $E[Y] = e^{s^2/2} = 3.00$
  - The flat tails miss more than half of it
- Reading: exp is convex and unbounded, so the top of the distribution carries much of the mean
  - A grid that stops early, read with a flat tail, understates the mean
  - Test 2: mean forecast / mean RV is 1.02 for Chronos-2 (grid to 99%), 0.97 for Chronos-Bolt and 0.94 for TimesFM-2.5 (grids to 90%)
  - Consistent with a cut upper tail, though the forecasts themselves also differ
- Report the rule with the forecasts

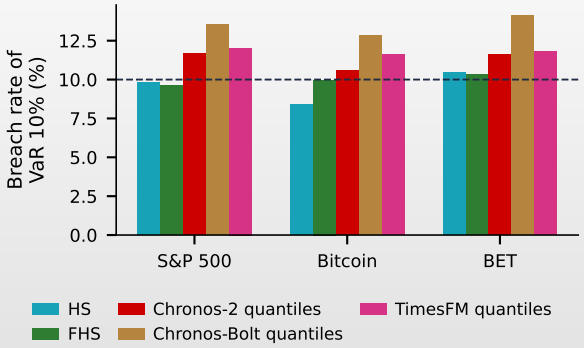
How Much Context Does Chronos-2 Need?



- QLIKE of Chronos-2 on log RV with contexts of 32 to 512 days, on the same forecast days
- QLIKE falls from 0.290 (32 days) to 0.231 (512 days)
  - Below about 256 days Chronos-2 is worse than log-HAR (0.234)
- Long memory: months of history reveal the slow-moving level
  - HAR encodes it explicitly through its 22-day average
- Context length is a modelling choice to report and test, like a window length

MFM_ch14_realised_vol

Where Deciles Suffice: VaR 10%



- Breach rate of the 10% quantile (target 10%) for HS, FHS and the three foundation models
- S&P 500: HS 9.8%, FHS 9.6%
  - Chronos-2 11.7%, Chronos-Bolt 13.5%, TimesFM-2.5 12.0%
- Kupiec rejects the foundation models in eight of nine cases
  - The lower tail sits too high, through location, scale or tail shape
- Plausible reason: pretraining corpora are dominated by smoother series
  - As volatility filters in FHS the models inherit the tails from the data

MFM_ch14_tail_risk

Recalibrating Quantiles: Adaptive Conformal Inference

- Guarantee for any sequence of returns, no model needed: over T days the breach rate approaches α

$$\left| \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1-\alpha_1) + \gamma}{\gamma T}$$

  - the bound shrinks like $1/T$: a long-run statement only, loose for small γ
- ES 2.5%: Chronos-2 ES shifted by the VaR 2.5% correction
  - A heuristic without a conformal guarantee
  - Rules and the full table in the Appendix

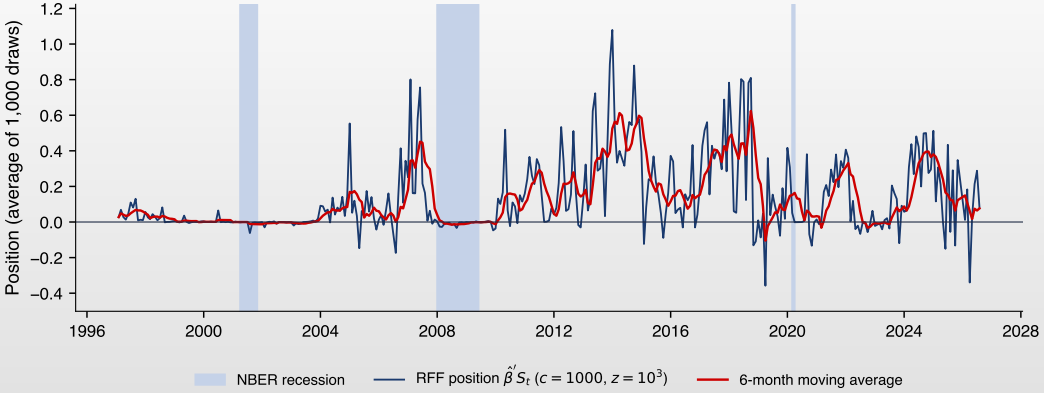
Case Study: Data and Method

- **Ridge** (Section II.A), no intercept

$$\hat{\beta}(z) = \left(zI + \frac{1}{T}\sum_t S_t S_t'\right)^{-1} \frac{1}{T}\sum_t S_t R_{t+1}$$

  - $z > 0$: ridge penalty, shrinks $\hat{\beta}$ towards zero; $z \to 0^+$: the ridgeless (pseudo-inverse) solution; T: window length
- **Design** (Section V.C): rolling windows of $T = 12$, 60, 120 months
  - P from 2 to 12 000; $\log_{10} z = -3, \ldots, 3$
  - Results averaged over 1 000 independent draws of the features
- **Evaluation**: out-of-sample $R^2 = 1 - \mathrm{Var}(\text{error})/\mathrm{Var}(R)$ (footnote 40)
  - The theory divides the MSE by $E[R^2]$, the zero forecast of Test 1
  - Annualised Sharpe ratio
  - alpha and information ratio (alpha over residual volatility) from a regression of the timing return on the market return
  - Benchmark: the linear "kitchen sink" regression on all 15 predictors (Section V.D)

Case Study: Timing Positions and Recessions (1/2)



MFM_ch14_virtue_complexity

Case Study: Timing Positions and Recessions (2/2)

- On course data, as in Kelly, Malamud & Zhou (2024): position $\hat{\beta}' S_t$ of RFF with $c = 1\,000$, $z = 10^3$
  - Shaded: NBER recessions (series `USREC` from FRED, Federal Reserve Economic Data)
  - Red: 6-month moving average
- Long in 72% of months, average position 0.13
  - A positive equity premium learned without a sign restriction
- Average position 0.07 in the 6 months before the 3 recessions and -0.01 during them
  - Against 0.15 otherwise
  - The 2020 pandemic shock is not anticipated
- Three recessions are a description, not a test: the paper has 15 recessions and 90 years

MFM_ch14_virtue_complexity

Pitfalls the Researcher Must Check

- **Leakage**: pretraining overlap, full-sample scaling, HAR averages that include day t
  - Seminar 14, exercise B10: they cut QLIKE by a quarter
  - Test: post-release window, simulated or private series
  - Recompute the forecast for t from data truncated at $t-1$
- **Weak benchmarks**: compare with zero (returns), log-HAR (volatility), FHS (tails), not with a random walk
- **Wrong functional**: medians used as means; deciles used as 1% quantiles
  - Mean forecast / mean RV should be near 1; Chronos-2 median: 0.79
- **Selection**: seeds, contexts and models chosen after seeing the test
  - Log every trial, compare the whole set with the MCS
- **Wrong inference**
  - DM on nested models
  - No multiplicity control
  - Coverage without independence
  - "Not significant" reported without the MDE

Norms, Spectral Radius and Explosion

- $\gamma_\varphi \|W\|_2 < 1$ is sufficient for geometric vanishing, so $\gamma_\varphi \|W\|_2 \geq 1$ is necessary for explosion (Pascanu, Mikolov & Bengio (2013))
- Spectral radius ρ (largest absolute eigenvalue)
  - In a linear RNN, $\rho(W) > 1$ is necessary for exponential explosion
  - $\rho(W) = 1$ with a Jordan block still gives polynomial growth
  - With φ the local Jacobians $D_t W$ matter: $\rho(W) = 0$ yet $\rho(DW) > 1$ is possible
- Stable VAR: impulse responses and gradients decay like $\rho(A)^k$
  - Up to a factor k^{m-1} if A has a Jordan block of size m
  - Jordan block: a repeated eigenvalue without a full set of eigenvectors
- DUx_t is an innovation only if the inputs x_t are centred white noise
  - Long-memory inputs would pass through the network

Back

Geometric versus Hyperbolic Decay, Fitted

- Constant gates, i.i.d. candidate input g_t, stationary start: $\mathrm{Corr}(c_t, c_{t-k}) = f^k$
- ARFIMA with $0 < d < 1/2$: $\rho_k \sim Ck^{2d-1}$
  - ARFIMA: autoregressive fractionally integrated moving average
- S&P 500 absolute returns since 2000: ACF 0.289 at lag 1, 0.132 at lag 50
  - Geometric decay with its own amplitude, $\rho_k = \rho_1 f^{k-1}$ (an AR(1) signal observed with noise), matches both lags with $f = 0.984$
  - $b_f = 4.1$, half-life 43 days
  - That f implies 0.027 at lag 150; the data show 0.059, more than twice as much
- A constant gate $f < 1$ is a short-memory filter
  - A long context or explicit long averages, as in HAR (Test 2), only approximate long memory over a finite range of lags

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Other Foundation Models and Their Corpora (1/2)

- Chronos-2: training corpus of selected Chronos and GIFT-Eval datasets
  - Plus synthetic multivariate series
- Benchmarks: GIFT-Eval (Aksu et al. (2024)) and fev-bench
- TimesFM corpus: Google Trends, Wikipedia page views, synthetic series
  - Also the M4 forecasting-competition series, electricity, traffic, weather
- Release dates of the weights, from the Hugging Face repositories
  - TimesFM-2.5: updated weight version announced on 15.09.2025, released on 01.10.2025
  - Chronos-2: announced on 20.10.2025, released on 30.10.2025

Back

Other Foundation Models and Their Corpora (2/2)

- **Moirai** (Woo et al. (2024)): one Transformer for any frequency and any number of variables
- The output is a mixture of distributions

$$p(y) = \sum_{k=1}^{4} w_k \, p_k(y), \qquad w_k \geq 0, \quad \sum_{k=1}^{4} w_k = 1$$

  - p_k: Student-t, log-normal, negative binomial and low-variance Normal densities
  - The weights w_k and the component parameters are network outputs, so the shape adapts to the series
- Moirai 2.0: Liu et al. (2025)
- **Lag-Llama** (Rasul et al. (2023)): decoder-only model with lagged values as features and a Student-t output

Back

A Formal Contamination Protocol (1/2)

- Pre/post regression on the loss differential $d_t = L_t^{FM} - L_t^{bench}$, with HAC standard errors

$$d_t = \beta_0 + \beta_1 1\{t \geq \tau\} + u_t$$

  - L_t^{FM}, L_t^{bench}: losses of the foundation model and of the benchmark on day t
  - τ: release date of the weights (a common cutoff when several models are compared)
  - Memorisation predicts $\beta_1 > 0$; H_0: $\beta_1 = 0$

- Report the minimum detectable effect (MDE) of β_1 (AI section)
  - Its floor 0.0169 uses the long-run variance of d_t in each period
  - The pre-release days, which include the April 2025 shock, are about 3.1 times as variable as the later ones
  - With one common variance the floor would be 0.0116

Back

A Formal Contamination Protocol (2/2)

- Pre-registration: forecasts submitted before the outcome exists make contamination impossible by design
  - Live benchmarks: TS-Arena (Meyer et al. (2025)), Impermanent (Garza et al. (2026))
  - Our post-release window is a manual version of the same idea
- Ablations: removing the language-model part of LLM-based forecasters does not degrade accuracy (Tan et al. (2024))
  - It often improves it
- Electricity prices: Pan & Ezzat (2026) benchmark on two datasets to limit contamination
- Result: foundation models are competitive but do not consistently beat domain-specific methods

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Return Forecasts, Full Table

R^2 (%) [DM p]	Historical mean	AR(1)	LSTM (5 seeds)	Chronos-2	Chronos-Bolt	TimesFM-2.5
S&P 500	+0.15 [0.095]	+0.94 [0.639]	+0.93 [0.090]	−0.26 [0.418]	−1.57 [0.078]	−3.35 [0.001]
Bitcoin	+0.07 [0.691]	−0.28 [0.326]	+0.04 [0.243]	−0.35 [0.413]	−0.61 [0.070]	−2.29 [0.001]
BET	+0.30 [0.081]	−0.00 [1.000]	+1.15 [0.014]	−0.39 [0.562]	+0.12 [0.831]	−3.27 [0.070]

- Point forecasts: grid medians (grid means: Design of the Return Test)
- DM: two-sided test on squared errors against the zero forecast, Newey–West variance

Back

Nested Forecasts, the BET AR(1) and Rolling Windows

- BET AR(1), $R^2 = -0.00\%$ and DM $p = 1.000$, is not a degenerate forecast
  - S.d. 0.074%, average slope 0.046
  - Squared-error gains of -12.7 in 2020 and $+14.7$ in 2026 cancel
  - Without these two years $R^2 = -0.11\%$
- Rolling or re-trained models: Giacomini & White (2006) test the forecasting method, estimation window included
  - Diebold (2015) on the use and abuse of DM
- SPA refines White's Reality Check (White (2000))
  - Very poor models (TimesFM) no longer dilute the test
  - Both use a stationary bootstrap with mean block 20 days

Back

Multiple Testing (1/2): Holm

- **Holm**: order the M p-values, $p_{(1)} \leq \cdots \leq p_{(M)}$
  - Reject while $p_{(j)} < 0.05/(M - j + 1)$, stop at the first non-rejection
- The adjusted $p_{(j)}$ is $\max_{i \leq j} \min\{1, (M - i + 1)\, p_{(i)}\}$
  - Toy example, $M = 3$, $p = 0.01, 0.02, 0.04$
  - Thresholds $0.05/3 = 0.0167$, $0.05/2 = 0.025$, 0.05: all three rejected
  - Bonferroni (0.0167 for all) rejects only the first
  - Adjusted p: 0.03, 0.04, 0.04
  - Chapter: the BET LSTM has the smallest of the 18 one-sided DM p-values, 0.0072
  - So its adjusted p is $18 \times 0.0072 = 0.130$

Back

Multiple Testing (2/2): Romano–Wolf and SPA

- **Romano–Wolf**: statistics $t_k = \sqrt{T}\,\bar{d}_k/\hat{\omega}_k$, ordered $t_{(\mathbf{1})} \geq \cdots \geq t_{(K)}$

$$p^{RW}_{(j)} = \max_{i \leq j} P^*\Big(\max_{k \in \mathcal{A}_i} t^*_k \geq t_{(i)} \Big), \qquad t^*_k = \frac{\sqrt{T}\,(\bar{d}^*_k - \bar{d}_k)}{\hat{\omega}_k}$$

  - With $\bar{d}_k$ the mean loss gain of model k over the zero forecast and $\hat{\omega}_k$ its HAC standard deviation
  - $\mathcal{A}_i$: the hypotheses not yet rejected at step i, i.e. $(i), \ldots, (K)$
  - P^*, $\bar{d}^*_k$: probability and mean under the stationary bootstrap
  - The threshold comes from the maximum of correlated statistics, so the procedure is less conservative than Holm for forecasts of one market
- **SPA**: one joint test of H_0: no model has a lower expected loss than the benchmark
  - Statistic $\max\left(0, \max_k t_k\right)$
  - p-value from a stationary bootstrap
  - Much worse models ($t_k \leq -\sqrt{2 \ln \ln T}$) are not recentred at zero, so they do not dilute the test
- Which question each answers
  - Holm and Romano–Wolf: *which* comparisons are significant with at most a 5% chance of any false rejection
  - SPA: *whether any* model beats the benchmark

Back

QLIKE, DM and MCS, Full Table

Model	QLIKE	DM vs log-HAR	p	MCS p	QLIKE after 3.11.2025
Random walk	0.326	+4.53	<0.001	0.004	0.384
HAR	0.247	+1.30	0.195	0.29	0.270
log-HAR	0.234	–	–	0.95	0.238
GARCH-t	0.283	+3.43	<0.001	0.03	0.295
LSTM	0.284	+1.68	0.094	0.29	0.233
Chronos-2	0.231	−0.62	0.537	1.00	0.242
Chronos-Bolt	0.232	−0.18	0.857	0.95	0.236
TimesFM-2.5	0.233	−0.08	0.938	0.95	0.244

- Literature: Brini (2026) compares nine zero-shot foundation models with eight HAR-type models on 50 assets
  - Pooled losses favour foundation models because of a few outlier assets
  - While asset by asset only one small model beats log-HAR at every horizon, narrowly
- Goel et al. (2025): TimesFM zero-shot is a reasonable baseline
  - Incremental fine-tuning is needed to beat econometric models

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VaR 1% Breaches and Kupiec Tests, Full Table

Breaches [Kupiec p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (26.9)	37 [0.065]	44 [0.002]	30 [0.559]	45 [0.001]	25 [0.705]	27 [0.989]
Bitcoin (28.2)	19 [0.065]	27 [0.822]	25 [0.539]	33 [0.374]	27 [0.822]	27 [0.822]
BET (26.8)	32 [0.322]	41 [0.010]	30 [0.536]	40 [0.016]	26 [0.884]	32 [0.322]

- In brackets after the market: the expected number of breaches, 1% of the days
- Kupiec: likelihood-ratio test of a breach rate $\alpha = 1\%$, with x breaches in T days and $\hat{\pi} = x/T$

$$\mathrm{LR}_{uc} = -2 \ln \frac{(1-\alpha)^{T-x}\alpha^{x}}{(1-\hat{\pi})^{T-x}\hat{\pi}^{x}} \sim \chi^{2}_{1}$$

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COVID-19 Crash, Scaling and Overshoot

- Chronos-2 standardises the context by its mean and standard deviation, then applies $\sinh^{-1}$ (Ansari et al. (2025))
  - On 16.03.2020 the crash raised the 512-day standard deviation by 9.7%
  - Mean absolute scale of Chronos: 3.2% (Seminar 14, exercise A6)
  - So the widening mixes the scale change and the reading of the crash days
  - Only an ablation could separate them
- Not a unit error: after the crash day Chronos-2 forecast a mean $|r|$ of 12.5% for the next day
  - Then FHS multiplies it by the 1% quantile of $r/\hat{\sigma}$ (4.7)
  - The filter extrapolates a one-day spike as a new level
  - VaR 4.6 times the largest loss, a high capital cost that the FZ0 loss penalises

Back

Berkowitz Tests and Quantile Scores, Three Markets

Market	Berkowitz p: Chronos-2	FHS	Below $q_{1\%}$ (C2)	QS: C2 / FHS	DM [p]
S&P 500	0.019	0.599	1.67%	0.510 / 0.506	+3.06 [0.002]
Bitcoin	0.595	0.306	1.17%	1.526 / 1.529	−0.90 [0.369]
BET	0.045	0.436	1.50%	0.465 / 0.460	+3.48 [<0.001]

- The Berkowitz test evaluates the linearly interpolated grid distribution function
  - And only μ, σ, not serial independence of the PIT
- QS weights the 21 levels equally
  - The CRPS would weight each level by its spacing
  - C2: Chronos-2; tie on Bitcoin

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FZ0 Losses and MCS p-Values, Full Table

FZ0 [MCS p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500	1.309 [0.14]	1.026 [0.19]	0.999 [1.00]	1.147 [0.14]	1.084 [0.14]	1.059 [0.18]
Bitcoin	2.237 [0.34]	2.192 [0.60]	2.192 [0.60]	2.181 [0.64]	2.151 [1.00]	2.160 [0.70]
BET	1.230 [0.015]	1.001 [0.34]	0.963 [1.00]	1.032 [0.24]	1.007 [0.34]	1.027 [0.20]

- Bitcoin: the hybrid FHS with Chronos-2 volatility has the lowest FZ0 loss
  - But the differences are not significant
- FZ0 for (VaR 2.5%, ES 2.5%)
  - The MCS uses the $T_{\max}$ statistic and a block bootstrap

Back

Conformal Recalibration, Rules and Results

After 250 calibration days: breaches [cc p / DQ p]	FHS	Chronos-2 raw	FHS + C2 vol.	Chronos-2 + ACI
S&P 500 (24.4)	28 [0.097 / 0.014]	42 [0.002 / <0.001]	23 [0.058 / <0.001]	21 [0.307 / 0.049]
Bitcoin (25.7)	24 [0.449 / 0.108]	31 [0.410 / 0.026]	26 [0.087 / 0.004]	24 [0.449 / 0.191]
BET (24.2)	26 [0.709 / 0.727]	36 [0.048 / 0.184]	23 [0.776 / 0.011]	25 [0.762 / 0.032]
FZ0, S&P 500 [DM p]	1.013	1.142 [0.009]	1.087 [0.022]	1.205 [0.001]

- $\hat{Q}$: the $\lceil (1-\alpha_t)(250+1) \rceil$-th smallest past score ($\lceil \cdot \rceil$: rounding up)
  - $+\infty$ if that rank exceeds 250 or $\alpha_t \leq 0$, $-\infty$ if $\alpha_t \geq 1$
  - With $\gamma = 0.005$: infinite VaR on 24 (S&P 500), 100 (Bitcoin), 270 (BET) days
- Guarantee with $T = 2\,443$, $\gamma = 0.001$: bound 0.41, uninformative for a 1% rate
  - Without exchangeability, weighted conformal bounds the gap by the drift (Barber et al. (2023))
- ES can be evaluated only jointly with VaR, through losses consistent for the pair such as FZ0 (Nolde & Ziegel (2017))
  - ES comparisons face finite-sample precision limits (Pele & Mazurencu-Marinescu-Pele (2026))
  - A GPD tail (McNeil & Frey (2000)) is an alternative for ES

Back

Case Study, Replication with Course Data

- Monthly SPY total returns from 1993 (course data) in excess of the 3-month Treasury bill rate
  - Treasury bill series: FRED `TB3MS`
  - Predictors of Welch & Goyal (2008) available: Treasury-bill rate (`tbl`), long-term yield (`lty`), term spread (`tms`), default yield spread (`dfy`), inflation (`infl`), all from FRED; stock variance (`svar`) from daily S&P 500 returns; the lagged excess return: 7 of the 15
  - Data limitation: the valuation ratios (dividend-price `dp`, dividend yield `dy`, earnings-price `ep`, payout `de`, book-to-market `b/m`), net equity expansion (`ntis`) and the bond returns (long-term return `ltr`, default return spread `dfr`) are not in the course data; about 30 years instead of 90
  - Method unchanged: $\gamma = 2$, $T = 12$, $P \leq 12\,000$, $\log_{\mathbf{10}} z = -3, \ldots, 3$, $1\,000$ draws
- Course-data results
  - Linear kitchen sink, $z = 0^+$ ($c = 7/12$): $R^{\mathbf{2}} = -1\,536\%$, Sharpe -0.06
  - Linear, $z = 10^{\mathbf{3}}$: $R^{\mathbf{2}} = 0.25\%$, information ratio 0.24 ($t = 1.3$) against SPY
  - RFF, $c = 1\,000$, $z = 10^{\mathbf{3}}$: $R^{\mathbf{2}} = -6.7\%$, information ratio -0.17 ($t = -0.9$) against SPY and 0.03 ($t = 0.1$) against the linear model

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Monthly Alignment and Missing Inflation

- Each predictor month forecasts the next calendar-month return
- A missing predictor does not move its target to a later month
  - Lags and targets keep the monthly calendar
- The consumer price index (FRED series CPIAUCSL) has no October 2025 observation in this data release
  - The missing value is not imputed
- The test targets run from February 1997 to August 2026: 353 forecasts
  - Except December 2025 and January 2026
- Volatility scaling uses 12 calendar months, before incomplete rows are removed
- The regression uses the latest 12 complete observations
  - Around missing data this window spans more than 12 months

The Discovery Loop, Reusable Prompts (1/2)

- **Literature**: `"Find evaluations of time-series foundation models on financial returns or volatility since 2024: models, data, loss, benchmark, and whether the test period overlaps the pretraining corpus."`
  - Semantic Scholar, Google Scholar, Elicit help with the search
- **Hypotheses**: `"List predictions that separate memorisation from skill for a zero-shot volatility forecaster: pre vs post release, never-public series, simulated series with the same dynamics."`
  - Diagnostics, not proof: a simulated log-HAR path tests transfer to that process
  - A pre/post gap can also reflect a regime change
- **Data and code**: `"From the daily SPY realised-variance forecasts of Chronos-2 and log-HAR, compute the QLIKE differential Chronos-2 minus log-HAR, test pre vs post 3 November 2025 with HAC errors, and report the minimum detectable effect at 80% power."`
  - Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**: `"Design a pre-registered test with more power: which cross-section of series, which variance proxy, which panel DM test, and how many post-release days."`
- **Critique**: `"Attack this claim: no memorisation, because the post-release DM test is insignificant. What is wrong with the inference?"`
  - Expect: low power, absence of evidence, and different regimes
  - The April 2025 tariff shock lies in the pre-release sample, the post-release window is calmer
- **Report**
  - Model versions and weight release dates
  - Context length, mean or median
  - Every seed and window tried

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Further Reading

- Gu, Kelly & Xiu (2020): machine learning for the cross-section of expected returns
- Kelly, Malamud & Zhou (2024): why very large models can predict returns better
- Ansari et al. (2024): Chronos, time series read as a language
- Crypto applications
  - Liquidity forecasting Agakishiev, Härdle & Pele (2025)
  - Early warning for "zombie" crypto-assets Będowska-Sójka, Wójcik & Pele (2026)
  - Statistical classification of crypto-assets Pele et al. (2023)
- Textbooks: Franke, Härdle & Hafner (2019) and Härdle, Chen & Overbeck (2017)
  - For the statistics of financial markets behind the benchmarks

Back

Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **ACI**: Adaptive Conformal Inference
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **CI**: Confidence Interval
- **COVID-19**: COronaVIrus Disease 2019
- **CPU**: Central Processing Unit
- **CRPS**: Continuous Ranked Probability Score
- **CRSP**: Center for Research in Security Prices
- **DM**: Diebold–Mariano (test)
- **DQ**: Dynamic Quantile (test)
- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund
- **FHS**: Filtered Historical Simulation
- **FM**: Foundation Model
- **FRED**: Federal Reserve Economic Data
- **FRTB**: Fundamental Review of the Trading Book
- **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GIFT**: General Time Series Forecasting Model Evaluation (benchmark GIFT-Eval)
- **GPD**: Generalised Pareto Distribution

Acronyms used in this material (2/2)

- **GRU**: Gated Recurrent Unit
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HS**: Historical Simulation
- **i.i.d.**: independent and identically distributed
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **LLM**: Large Language Model
- **LN**: Layer Normalisation
- **LR**: Likelihood Ratio (test)
- **LSTM**: Long Short-Term Memory (network)
- **MCS**: Model Confidence Set
- **MDE**: Minimum Detectable Effect
- **MHA**: Multi-Head Attention
- **MLP**: MultiLayer Perceptron
- **MSE**: Mean Squared Error
- **NBER**: National Bureau of Economic Research
- **PIT**: Probability Integral Transform
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QS**: Quantile Score
- **ReLU**: Rectified Linear Unit
- **RFF**: Random Fourier Features
- **RNN**: Recurrent Neural Network
- **RV**: Realised Variance
- **SPA**: Superior Predictive Ability (test)
- **SSE**: Sum of Squared Errors
- **T5**: Text-to-Text Transfer Transformer
- **TS**: Time Series (in the name TS-Arena)
- **US**: United States
- **VAR**: Vector AutoRegression

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